

JAIBB

Hand Notes

On

**Monetary And Financial System
(MAFS)**

Module	SL	Topics	Page
A	1	What is money? List and summarize the primary functions of money. 98(1.a)	5
	2	Write down Keynesian money demand function and explain. 98(2.a)	6
	3	Why do people have a demand for money? Which motive of demand for money is influenced by income and which by rate of interest? 97(3.b)	7
	4	What is money supply? Does money supply includes interbank deposits? Why? 97(2.a)	8
	5	How money supply is measured? Compare and contrast M1 and M2 money supply. 96(1.a)	8
	6	Clarify 2 (two) determinants of money supply. Can monetary authority/central bank directly control money supply? Explain.96(1.c)	10
	7	If you use a 'credit card' to purchase goods or services on the internet, does this affect the M1 or M2 money supply or both or neither? Explain. 96(3.c)	10
	8	Briefly discuss the commercial banks role in the process of the money supply. 96(4.a)	11
	9	Draw a relation between nominal rate of interest, real rate of interest and inflation. 98(2.c)	12
	10	Define 'real' and 'nominal' interest rate in brief. Is real interest rate can be negative at the time of stagflation? Explain with example.97(6.c)	12
	11	What are the constituents of monetary system? 96(4.a)	13
	12	Explain how banks create money through multiple deposit creation model. Assume that reserve requirement is 10% and initial deposit is Tk. 100. 99(2.a)	14
	13	How banks do create money? If central bank sells Government Treasury Bond through auctions, what will be the impact of this auction on money supply? 97(3.c)	15
B	1	Describe different Payment System in bangladesh.	17
	2	Describe the most common types EFT payments in bangladesh.	18
	3	Advantages and Disadvantages of Different Payment Methods.	19
	4	Why electronic payments is popular? Discuss briefly. 98(3.a)	21
	5	Which payment method i.e. MPS/Traditional is more effective to enhance wage earner remittance? Explain. 98(3.c)	22
	6	What do you mean by contactless payment? Explain. 98(3.d)	23
	7	What challenges banks are facing in implementing electronic payments in Bangladesh? Describe briefly. 97(4.a)	24
	8	Which payment method, ie, cash, debit/credit cards, MFS, EFT, QR, is the most dominating and which one is the fastest growing in Bangladesh? Explain briefly. 97(5.c)	25
	9	Distinguish between 'debit card' and 'credit card'. Describe pros and cons of using a cheque as a payment method. 96(3.a)	25
C	1	Different forms of Finance (Rudimentary, Direct and Indirect).	27
	2	What is a financial system? Which components constitute a financial system? 96(5.a)	28
	3	Differentiate between monetary system, financial system and payment system of a country. 99(3.a)	30
	4	Describe the financial system of Bangladesh. 99(3.b)	30
	5	Name the main regulators of the financial system of Bangladesh. Identify the roles of BSEC and BB to develop a vibrant securities market in Bangladesh. 99(6.a)	32
	6	"All forms of money are financial instruments but all financial instruments are not money". Explain. 97(6.a)	33
	7	What are characteristics of efficient financial market? 98(5.c)	34
	8	Money market Vs Capital market	35

	9	Explain financial development in brief. How you can measure the financial development of a country? Illustrate with 3 different methods. 96(5.b)	36
	10	Give example of five financial development indicators. 99(6.b)	37
	11	What are differences between micro credit and micro finance? 98(4.d)	37
	12	Explain the development role of finance.	38
	13	How transaction costs influence Financial System?	38
	14	What are the key functions performed by financial systems? Explain in brief.	39
	15	Difference between Spot, Future, Forward and Option markets.	39
	16	Difference between Primary and Secondary Market.	40
D	1	Explain how Banking Financial Institutions (BFIs) and Non-bank Financial Institutions (NOFIs) serve as financial intermediaries. 96(7.a)	41
	2	Outline the functions and growth of BFIs in Bangladesh. 96(7.b)	42
	3	Outline the functions and growth of NBFIs in Bangladesh. 96(7.b)	43
	4	What are the different roles banks play in an economy?	44
	5	What are the different aspects of Asset-Liability Management (ALM)?	44
	6	What are the key trends in commercial banking?	45
E	1	What is the purpose of the money markets?	46
	2	Who are the participants in the money markets and what are their roles?	46
	3	What are the key instruments used in the money markets?	47
	4	What is the capital market and what are its functions?	48
	5	Who are the participants in the capital market?	48
	6	What is the bond market and how does it operate?	49
	7	What is the distinction between interest rates and returns in the financial markets?	50
F	1	What are the Islamic governing principles (Shariah) in finance?	51
	2	What do you know about Shari'ah governance in an Islamic financial institution? Outline the key players and their role in a 'Shari'ah governance framework'. 96(9.c)	51
	3	What is the relationship between Finance and Religion (Islam)?	52
	4	Briefly explain the basic fundamentals of Shari'ah Law? List major principles that must be followed while developing Shari'ah compliant products. 96(9.a)	53
	5	How come Ijma & Qiyas (as source of Shariah law) are different from Quran and Hadith? 98(9.d)	54
	6	Explain the role of money in Islamic economics and finance. 98(9.a)	55
	7	Distinguishing features of Islamic Banking	56
	8	Briefly explain the objectives of Islamic Banking. How Islam interprets 'Riba'?	56
	9	What are the key principles of Islamic banking? What are the conceptual differences of 'interest' and 'riba'? 97(7.a)	57
	10	Based on principles, what are the available Islamic financial instruments? How do you differentiate between them? 98(9.c)	58
	11	List three key Shariah Principles to adopted by Islamic banks in delivering their products. 98(9.b)	59
	12	What are the investment products of Islamic banks? Explain the key features of the investment products of Islamic banks in Bangladesh State and explain the fundamental sources of Shariah Law. 97(7.b)	59
	13	What are the major prohibitions that must be followed while developing a shariah compliant financial services model? How sukuk is different from commercial bonds?	60
	1	Why is financial system regulation necessary?	62
	2	What are the different types of financial system regulations?	62

G	3	What are the key objectives of the central bank?	63
	4	What are the main functions of the central bank?	63
	5	What are the methods of credit control used by the central bank?	64
	6	Who are the regulators of the financial system in Bangladesh?	65
	7	What are the core policies of the central bank?	65
	8	Distinguish between 'Macroprudential' and 'Microprudential' regulations. 97(1.c)	66
		Short Notes	67 – 69
		Mathematical Problem	70 - 87

Module-A: Money and Monetary System

1. What is money? List and summarize the primary functions of money. 98(1.a)

Money:

Money is anything that is generally accepted as a medium of exchange for goods and services and as a means of payment for debts. The essential feature of money is its general acceptability in an economy, which is based on the collective belief that it will be accepted by everyone.

Primary Functions of Money

1. Medium of Exchange

- ✓ The most important function of money is to facilitate transactions by eliminating the need for barter.
- ✓ It removes the problem of the double coincidence of wants, making trade efficient.
- ✓ In Bangladesh, the taka (BDT) serves as the official medium of exchange.

2. Unit of Account

- ✓ Money provides a common measure to express the value of goods and services.
- ✓ It allows easy comparison of prices and efficient economic decision-making.
- ✓ In Bangladesh, prices of products and services are quoted in taka.

3. Standard of Deferred Payment

- ✓ Money is used for contracts involving future payments, such as loans, salaries, and rents.
- ✓ It enables the development of financial markets and credit systems.
- ✓ Banks in Bangladesh use taka for loan agreements and installment payments.

4. Store of Value

- ✓ Money retains its value over time, allowing people to save and store wealth.
- ✓ Though inflation can erode its value, money remains more liquid than other assets.
- ✓ Bangladesh Bank regulates money supply to maintain stability in its value.

These functions make money essential for a stable and efficient monetary system.

2. Write down Keynesian money demand function and explain. 98(2.a)

Keynesian Money Demand Function

John Maynard Keynes proposed the **Liquidity Preference Theory**, which explains why individuals hold money. According to Keynes, the demand for money (**Md**) depends on **income (Y)** and **interest rate (i)** and is expressed as:

$$Md = L(Y, i)$$

Where:

- Md = Demand for money
- $L(Y, i)$ = Liquidity preference function
- Y = Income (positively related to money demand)
- i = Interest rate (negatively related to money demand)

Explanation of Keynesian Money Demand

Keynes identified three motives for holding money:

1. Transaction Motive

- ✓ People hold money to facilitate daily transactions and purchases.
- ✓ Demand for transaction money increases with income.
- ✓ In Bangladesh, businesses and individuals hold money to cover expenses like rent, salaries, and utility bills.

2. Precautionary Motive

- ✓ Money is held as a precaution for unexpected expenses (e.g., medical emergencies or urgent repairs).
- ✓ This demand is also proportional to income.
- ✓ In Bangladesh, families often keep savings in cash for sudden medical needs or financial emergencies.

3. Speculative Motive

- ✓ People hold money to take advantage of future investment opportunities.
- ✓ When interest rates are **low**, people hold more cash; when interest rates **rise**, they invest in bonds or other assets.
- ✓ In Bangladesh, investors decide whether to hold cash or invest in government securities based on interest rate movements.

Keynes' theory highlights that money demand depends on both income and interest rates. While income influences the **transaction and precautionary** motives, interest rates affect the **speculative** motive. In Bangladesh, the central bank (Bangladesh Bank) monitors money demand to manage inflation and financial stability.

3. Why do people have a demand for money? Which motive of demand for money is influenced by income and which by rate of interest? 97(3.b)

Why Do People Have a Demand for Money:

People demand money for several reasons, which can be categorized into three main motives:

1. **Transactions Motive:**
People hold money to carry out everyday transactions, such as buying goods and services. Money is used as a medium of exchange to facilitate the buying and selling process.
2. **Precautionary Motive:**
Individuals hold money as a safeguard against unforeseen circumstances, like unexpected medical expenses or emergencies. It acts as a buffer to cover future uncertain needs.
3. **Speculative Motive:**
People hold money as a store of wealth in order to take advantage of future investment opportunities or avoid losing value in other assets. This motive is influenced by the rate of interest and the expectation of future changes in prices or interest rates.

Motive Influenced by Income and Rate of Interest

- **Income and Transactions Motive:**
The **transactions motive** is primarily influenced by a person's income. As income increases, people are likely to demand more money for daily transactions because they can afford to engage in more transactions.
- **Rate of Interest and Speculative Motive:**
The **speculative motive** is influenced by the rate of interest. When interest rates are high, people are less likely to hold money as they can earn more from investments (bonds, stocks, etc.). Conversely, when interest rates are low, people are more likely to hold onto money because the opportunity cost of holding cash is lower.

Transactions motive is influenced by income, while the **speculative motive** is influenced by the rate of interest.

4. What is money supply? Does money supply includes interbank deposits? Why? 97(2.a)

Money Supply:

Money supply refers to the total amount of money circulating in an economy at a given point in time. It includes all forms of money that are available for spending and investment. The money supply is measured in different ways to capture various aspects of liquidity in the economy. These measures include:

1. **M1 (Narrow Money):** Includes currency in circulation outside the banking system (Currency Outside Banks - COB) and demand deposits (DD) held by the public.
2. **M2 (Broad Money):** Includes M1 plus time deposits held by the public in banks.

Money Supply Include Interbank Deposits:

Money supply does not include interbank deposits. Because-

- **Interbank deposits** are deposits held by one bank in another bank. These deposits are part of the banking system's reserves and are not available for public use or consumption.
- Money supply measures focus on the money that is available to the general public for spending or investment. Since interbank deposits are held between financial institutions, they do not directly influence the purchasing power or liquidity available to the public.
- Including interbank deposits in money supply would artificially inflate the amount of money available to the public, leading to misleading conclusions about the economy's liquidity.

Interbank deposits are excluded from the money supply because they are not part of the money that is accessible or spent by households and businesses.

5. How money supply is measured? Compare and contrast M1 and M2 money supply.

How is Money Supply Measured:

Money supply is typically measured in terms of the total quantity of money circulating in an economy. It includes different components based on their liquidity, or how easily they can be converted into cash or used for transactions. The central bank of a country typically measures money supply to understand economic conditions and manage monetary policy. The common measures of money supply are:

- **M1:** Narrow money, which includes the most liquid forms of money.
- **M2:** Broader measure, which includes M1 plus less liquid assets.

Comparison of M1 and M2 Money Supply

Measure	M1 Money Supply	M2 Money Supply
Definition	M1 includes the most liquid forms of money: currency in circulation outside banks (COB) and demand deposits (DD).	M2 includes all of M1 plus time deposits (TD) and certain other types of deposits that are less liquid.
Components	- Currency Outside Banks (COB) - Demand Deposits (DD)	- M1 (Currency Outside Banks + Demand Deposits) - Time Deposits (TD)
Liquidity	Highly liquid – used directly for transactions.	Less liquid than M1 – time deposits have a fixed term and can't be accessed without penalty.
Function	Used for immediate transactions and purchases.	Includes savings and less liquid assets, providing a broader view of the money available in the economy.
Impact on Economy	Reflects the money readily available for consumption and short-term spending.	Offers a more comprehensive view of money supply that includes savings and less accessible funds, indicating future spending potential.

Key Differences:

1. Liquidity:

- ✓ **M1** is the most liquid form of money, meaning it is available for immediate use in transactions.
- ✓ **M2** is broader and includes less liquid forms of money, like time deposits that require waiting to access.

2. Composition:

- ✓ **M1** is limited to currency and demand deposits, which are most easily used for day-to-day purchases.
- ✓ **M2** includes everything in M1, plus additional forms like time deposits, which cannot be quickly converted into cash without a penalty.

3. Economic Indicator:

- ✓ **M1** reflects the current spending power in the economy, and its changes can indicate changes in consumer confidence and economic activity.

- ✓ **M2** gives a broader view of money available in the economy, indicating future consumption and investment trends.

6. Clarify 2 (two) determinants of money supply. Can monetary authority/central bank directly control money supply? Explain. 96(1.c)

Two Determinants of Money Supply:

1. **Central Bank's Monetary Policy:** The central bank controls the money supply through tools like **open market operations**, **reserve requirements**, and the **discount rate**. By buying/selling government securities or adjusting reserve requirements, the central bank can increase or decrease the amount of money circulating in the economy.
2. **Banking System's Lending Behavior:** The commercial banks' ability to create money through lending affects the money supply. When banks lend more, they create deposits, expanding the money supply. Conversely, if banks reduce lending, the money supply contracts.

Can the Central Bank Directly Control Money Supply?

No, the **central bank** cannot directly control the money supply. It can influence it through policy tools, but other factors like **public demand for money** (how much money people hold) and **banks' lending behavior** impact the money supply. For instance, if the public chooses to hold more cash or if banks tighten lending, it can limit the central bank's ability to control money supply precisely. Therefore, while the central bank can influence, it does not have full control.

7. If you use a 'credit card' to purchase goods or services on the internet, does this affect the M1 or M2 money supply or both or neither? Explain. 96(3.c)

When you use a **credit card** to purchase goods or services on the internet, it does not directly affect the **M1** or **M2** money supply at the moment of the transaction.

M1 Money Supply:

- **M1** includes currency in circulation and demand deposits (like checking accounts), which are immediately accessible for spending.
- When you use a **credit card**, you are borrowing money from the credit card issuer (typically a bank), not using your own deposit or currency. The transaction itself does not involve transferring funds from your demand deposit (checking account) or currency.

- Therefore, using a credit card does **not** directly affect the **M1** money supply.

M2 Money Supply:

- **M2** includes **M1** plus time deposits, savings accounts, and other near-money assets.
- Using a credit card still does not directly change the money in circulation or deposits in savings accounts, so it does **not** immediately affect **M2** either.

Using a credit card to purchase goods or services does **not directly** affect either **M1** or **M2** at the time of the transaction, but it can have an indirect effect when the credit card bill is paid and funds are transferred from your accounts.

8. Briefly discuss the commercial banks role in the process of the money supply. 96(4.a)

Commercial Banks' Role in the Money Supply Process

Commercial banks play a significant role in money creation by lending out a portion of the deposits they receive, thus increasing the money supply. Here's a brief overview:

1. **Deposits and Reserves:** When customers deposit money, banks are required to keep a fraction (reserve requirement) of that deposit as reserves and lend out the remaining amount.
2. **Lending and Money Creation:** The loans provided by banks are redeposited and may be lent out again, creating more money in the economy. This process is called money creation.
3. **Money Multiplier:** The total money supply in the economy can increase by a multiple of the initial deposit, depending on the reserve requirement ratio. For example, with a 10% reserve requirement, the money multiplier is 10.
4. **Impact on Money Supply:** Through lending, commercial banks can significantly expand the money supply, contributing to economic growth.

Commercial banks create money by lending out deposits, and their lending behavior, combined with the reserve requirement, determines the total money supply.

9. Draw a relation between nominal rate of interest, real rate of interest and inflation.

The relationship between the **nominal rate of interest**, **real rate of interest**, and **inflation** is typically expressed by the **Fisher equation**. This equation shows how inflation affects the nominal interest rate.

The Fisher equation is:

$$1 + i = (1 + r)(1 + \pi)$$

Where:

- i = Nominal interest rate
- r = Real interest rate
- π = Inflation rate

For small values of inflation, this can be approximated as:

$$i \approx r + \pi$$

If inflation increases, the nominal interest rate increases to maintain the real purchasing power of the return on investment.

The real interest rate is the return after accounting for inflation, while the nominal rate includes inflation.

10. Define 'real' and 'nominal' interest rate in brief. Is real interest rate can be negative at the time of stagflation? Explain with example. 97(6.c)

Nominal Interest Rate:

The **nominal interest rate** is the interest rate before adjusting for inflation. It is the rate quoted by financial institutions and represents the actual monetary return or cost of borrowing. Nominal interest rate does not account for changes in purchasing power due to inflation.

Real Interest Rate:

The **real interest rate** adjusts the nominal interest rate for inflation. It reflects the true purchasing power of the interest earned or paid, indicating how much the lender or borrower actually gains or loses in real terms after accounting for inflation. The real interest rate is calculated using the formula:

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

Can the Real Interest Rate be Negative During Stagflation:

Yes, the real interest rate can be negative during **stagflation**, which is a period of stagnant economic growth combined with high inflation and high unemployment.

Example:

Suppose the nominal interest rate is 5%, and inflation during stagflation is 8%. The real interest rate would be:

$$\text{Real Interest Rate} = 5\% - 8\% = -3\%$$

In this case, even though the nominal interest rate is positive, the real interest rate becomes negative because inflation exceeds the nominal rate. This situation means that the purchasing power of money decreases over time, and lenders lose in real terms.

Explanation:

- ✓ During stagflation, inflation is high, which erodes the value of money.
- ✓ A negative real interest rate means that lenders receive less value than they initially lent out, and borrowers effectively pay back less in real terms.
- ✓ For example, if a person borrows 100,000 Taka at 5% interest, but inflation is 8%, they repay the loan in nominal terms, but in real terms, they are effectively paying less due to the reduction in the purchasing power of money.

11. What are the constituents of monetary system? 96(4.a)

Constituents of the Monetary System:

The monetary system consists of the following key components:

1. **Currency:** Physical money, including coins and banknotes, issued by the central bank or government, used for transactions.
2. **Banks and Financial Institutions:** Commercial banks provide services like loans and deposits, while the central bank regulates money supply and financial stability.
3. **Money Supply:** The total amount of money in circulation, measured through aggregates like M1, M2 which include currency and deposits.

4. **Payment Systems:** Mechanisms for transferring money between individuals and institutions, such as electronic payments, credit/debit cards, and online transactions.

These elements together ensure the proper functioning of the monetary system in an economy.

12. Explain how banks create money through multiple deposit creation model. Assume that reserve requirement is 10% and initial deposit is Tk. 100. 99(2.a)

Banks Create Money through Multiple Deposit Creation Model

In the multiple deposit creation model, banks can create money through lending activities. This process depends on the reserve requirement and how the bank lends out a fraction of the deposit while keeping a portion as reserves. Here's an explanation with the assumptions:

- **Reserve Requirement (RR):** 10% (0.10)
- **Initial Deposit:** Tk. 100

Steps in the Money Creation Process:

1. **Initial Deposit:** A customer deposits Tk. 100 in the bank.
2. **Bank Keeps Reserves:** The bank must keep 10% of the deposit as reserves (according to the reserve requirement). Therefore, the bank keeps Tk. 10 as reserves.

$$\text{Reserve} = 100 \times 10\% = Tk.10$$

3. **Lending the Remaining Money:** The remaining amount, which is Tk. 90, can be lent out to borrowers.

$$\text{Loan Amount} = 100 - 10 = Tk.90$$

4. **The Borrowed Money is Deposited Again:** The borrower deposits the Tk. 90 loan into the bank (same or another bank).
5. **Bank Reserves 10%:** The new deposit of Tk. 90 requires the bank to keep 10% as reserves (Tk. 9). The bank can then lend out the remaining amount (Tk. 81).

$$\text{Reserve for New Deposit} = 90 \times 10\% = Tk.9$$

↓

$$\text{Loan Amount} = 90 - 9 = Tk.81$$

6. **Repeat the Process:** This process continues, with the bank keeping 10% of the new deposit as reserves and lending out the rest.

Money Multiplier:

The total money supply in the economy is determined by the **money multiplier**. The formula is:

$$\text{Money Multiplier} = \frac{1}{\text{Reserve Requirement}}$$

In this case, with a 10% reserve requirement:

$$\text{Money Multiplier} = \frac{1}{0.10} = 10$$

Total Money Created:

The total money created in the banking system can be calculated as:

$$\text{Total Money Supply} = \text{Initial Deposit} \times \text{Money Multiplier}$$

$$\text{Total Money Supply} = 100 \times 10 = \text{Tk.1,000}$$

Through multiple deposit creation, the initial deposit of Tk. 100 can lead to a total money supply of Tk. 1,000 in the economy. This process demonstrates how banks can effectively "create" money by lending out a portion of the deposits they receive, while maintaining the required reserves.

13. How banks do create money? If central bank sells Government Treasury Bond through auctions, what will be the impact of this auction on money supply? 97(3.c)

How Banks Create Money:

1. **Initial Deposit:** A customer deposits money into the bank.
2. **Reserve Requirement:** The bank keeps a fraction (e.g., 10%) as reserves and lends out the remaining amount.
3. **Money Creation:** The loaned money is redeposited in the banking system, allowing further lending and increasing the money supply.
4. **Money Multiplier:** The total money supply increase is determined by the money multiplier formula:

$$\text{Money Multiplier} = \frac{1}{\text{Reserve Requirement}}.$$

Impact of Treasury Bond Auction on Money Supply:

1. **Bond Sale:** The central bank sells government treasury bonds.
2. **Decrease in Reserves:** Banks or investors pay for the bonds, reducing the reserves in the banking system.
3. **Lower Lending Capacity:** With fewer reserves, banks can lend less, contracting the money supply.
4. **Money Supply Reduction:** The overall money supply in the economy decreases due to this reduced lending capacity.

Module-B: Payment System

1. Describe different Payment System in Bangladesh.

Different Payment Systems in Bangladesh

1. Cash Payment System:

- ✓ Traditional method of payment using physical currency (coins and banknotes).
- ✓ Common for small transactions, especially in rural areas.
- ✓ Cash payments are direct, easy to use, but less secure for large amounts.

2. Cheque Payment System:

- ✓ A cheque is a written order directing a bank to transfer money from one account to another.
- ✓ Common for business transactions and large payments.
- ✓ Provides a record of payment but takes time to clear and settle.

3. Mobile Financial Services (MFS):

- ✓ Mobile banking platforms like *bKash*, *Nagad*, and *Rocket* allow users to send money, pay bills, and perform other transactions via mobile phones.
- ✓ MFS has gained significant popularity due to its convenience, especially in rural areas.
- ✓ Widely used for peer-to-peer money transfers, utility payments, and merchant transactions.

4. Real-Time Gross Settlement (RTGS):

- ✓ RTGS is an electronic payment system for high-value and time-sensitive transactions.
- ✓ It operates in real-time, settling payments instantly between banks.
- ✓ Mainly used by financial institutions for interbank transactions and large corporate payments.

5. Electronic Funds Transfer (EFT):

- ✓ EFT allows the transfer of funds electronically between bank accounts.
- ✓ Services like *ATM*, *POS* (Point of Sale) terminals, and mobile banking use EFT to process payments.
- ✓ Used for transferring money within the same bank or across different banks.

6. Debit/Credit Card Payments:

- ✓ Debit and credit card payments are common for purchasing goods and services, especially in urban areas.
- ✓ Banks issue these cards, and users can make payments through *POS terminals* or online shopping platforms.
- ✓ Secure payment method, though not universally accepted in rural areas.

7. Online Payment Gateways:

- ✓ Payment systems like *SSLCommerz* and *EasyPay* enable online payments for e-commerce and digital services.
- ✓ These gateways facilitate payments through cards, bank transfers, and mobile wallets.
- ✓ Widely used for online shopping, bill payments, and subscription-based services.

8. Bank Transfer via Internet Banking:

- ✓ Internet banking services allow customers to transfer money between their own accounts or to other banks.
- ✓ Users can make payments for utility bills, loans, and other services directly from their accounts.

9. Payment through Post Offices:

- ✓ Post offices in Bangladesh also offer financial services, including money transfer, savings, and bill payment.
- ✓ *PostBank* provides an online payment facility for utility bills and banking services, especially for people in remote areas.

These payment systems have significantly evolved in Bangladesh, catering to the diverse needs of the population, and facilitating economic growth and financial inclusion.

2. Describe the most common types EFT payments in Bangladesh.

Most Common Types of EFT Payments in Bangladesh

1. Credit or Debit Card Payment:

- ✓ Used extensively for consumer purchases in retail outlets, online platforms, and bill payments.
- ✓ Allows for both domestic and international transactions, making it convenient for a wide range of consumers.

2. **Automated Clearing House (ACH) Payments:**

- ✓ Efficient for processing bulk payments such as payroll, loan repayments, and utility bills.
- ✓ Reduces paperwork and processing time, making transactions faster and cost-effective.

3. **eCheck:**

- ✓ Digital alternative to traditional paper checks, used for business-to-business payments and online transactions.
- ✓ Requires only the bank's routing number and account number, ensuring secure and simple transactions.

4. **Wire Transfers:**

- ✓ Used for transferring large sums of money, especially in international transactions.
- ✓ Offers a secure method for transferring high-value transactions with minimal risk and time delay.

3. Advantages and Disadvantages of Different Payment Methods.

Advantages and Disadvantages of Different Payment Methods

1. **Cash Payment**

Advantages:

- ✓ One of the most common and easiest forms of payment.
- ✓ Customers expect businesses to accept cash.

Disadvantages:

- ✓ Carrying large amounts of cash is not convenient for customers.
- ✓ Storing or transporting cash can be risky.

2. **Check Payment**

Advantages:

- ✓ Allows customers to make large purchases safely.
- ✓ You don't need to keep as much cash on hand in the store.

Disadvantages:

- ✓ There can be delays in processing checks, and funds may not be available immediately.
- ✓ Risk of receiving a bounced or fake check.

3. **Debit, Credit, and Prepaid Cards**

Advantages:

- ✓ Facilitates frequent and large purchases, safer than carrying cash.
- ✓ Quick and convenient payment method.

Disadvantages:

- Transactions may take 1-3 days to process.
- There may be transaction fees and equipment costs for accepting payments.

4. **Mobile Payments**

Advantages:

- ✓ Encourages frequent and larger purchases, convenient for both businesses and customers.
- ✓ Secure and reliable payment method, especially for international travelers.

Disadvantages:

- ✓ Payment processing may take 1-3 days.
- ✓ Transaction fees and the need for special devices to accept payments.

5. **Electronic Bank Transfers**

Advantages:

- ✓ Facilitates large payments and is secure.
- ✓ Customers can safely make large purchases without worrying about counterfeit money.

Disadvantages:

- ✓ Non-business customers may not feel comfortable transferring money directly.
- ✓ You may need to set up the transaction with both your and the customer's bank, which can be cumbersome.

6. **Mobile Wallet**

Advantages:

- ✓ More secure than using physical cards due to encrypted data.

- ✓ Fast checkout process, increasing the frequency of purchases.

Disadvantages:

- ✓ Requires a device to process the transaction (POS or mobile payment device).

7. **QR Code Payment**

Advantages:

- ✓ Contactless, offering convenience and hygiene.
- ✓ Works with all smartphones and does not require a specific app.

Disadvantages:

- ✓ A POS system or terminal is not needed for the transaction, but it may require internet access to function.

4. Why electronic payments is popular? Discuss briefly. 98(3.a)

Why Electronic Payments are Popular:

Electronic payments have gained popularity due to their convenience, security, and efficiency. The key reasons are:

1. **Speed and Convenience**

- Transactions are processed instantly, reducing the need for physical visits to banks.
- Payments can be made anytime, anywhere using mobile apps or online banking.

2. **Security and Transparency**

- Electronic payments reduce the risk of theft or fraud compared to cash transactions.
- Digital records provide transparency and make tracking transactions easier.

3. **Cost-Effectiveness**

- Businesses save money by reducing cash handling and processing costs.
- Customers can avoid extra charges related to cash withdrawals or check processing.

4. **Global Reach**

- Enables cross-border transactions without the hassle of currency exchange.
- Supports international trade and online shopping.

5. Supports Financial Inclusion

- People without access to traditional banks can use mobile banking or digital wallets.
- Encourages cashless transactions, boosting economic efficiency.

These factors make electronic payments a preferred choice for businesses and individuals worldwide.

5. Which payment method i.e. MPS/Traditional is more effective to enhance wage earner remittance? Explain. 98(3.c)

More Effective Payment Method for Enhancing Wage Earner Remittance: MPS vs. Traditional Methods

To enhance wage earner remittance, **Mobile Payment System (MPS)** is more effective than traditional methods due to its speed, accessibility, and cost-effectiveness. Here's a comparison:

1. Speed and Convenience

- **MPS:** Enables instant fund transfers through mobile banking and digital wallets (e.g., bKash, Nagad, Rocket), reducing processing time.
- **Traditional Methods:** Bank drafts or money orders take days or weeks to process, causing delays.

2. Accessibility and Inclusion

- **MPS:** Beneficiaries can receive funds without a bank account, making it ideal for rural areas.
- **Traditional Methods:** Requires recipients to visit a bank, which can be challenging in remote locations.

3. Cost-effectiveness

- **MPS:** Lower transaction fees, reducing remittance costs for wage earners.
- **Traditional Methods:** Higher fees due to bank and intermediary charges.

4. Security and Transparency

- **MPS:** Transactions are encrypted and provide instant confirmations, ensuring safety.
- **Traditional Methods:** Risk of fraud or delays in fund delivery.

5. Government and Policy Support

- The Bangladesh government promotes **digital remittance channels** through MFS and mobile banking, encouraging formal remittance inflows.

MPS is more effective than traditional methods for enhancing wage earner remittance due to its **speed, accessibility, lower costs, and security**. It facilitates financial inclusion and boosts formal remittance flows, benefiting both expatriates and their families.

6. What do you mean by contactless payment? Explain. 98(3.d)

Contactless Payment:

Contactless payment is a secure method of transaction where customers can make payments without physically swiping or inserting a card into a terminal. Instead, it uses **Near Field Communication (NFC)** or **Radio Frequency Identification (RFID)** technology to complete transactions by simply tapping or waving a payment card, smartphone, or wearable device near a compatible payment terminal.

How Contactless Payment Works:

1. **NFC Technology:** Contactless cards or mobile devices communicate with the payment terminal using NFC, enabling fast and secure transactions.
2. **Tap-and-Go:** Users just tap or hover their card, phone, or smartwatch over the payment machine.
3. **Authentication:** For small transactions, no PIN is needed; for larger amounts, authentication may be required.
4. **Instant Processing:** The transaction is completed in seconds, making it faster than traditional methods.

Advantages of Contactless Payment:

- **Faster Transactions:** Reduces waiting time at checkout.
- **More Secure:** Uses encryption and tokenization for fraud protection.
- **Convenient:** No need to carry cash or touch payment terminals.
- **Hygienic:** Reduces physical contact, beneficial during pandemics.

Example:

- **NFC-enabled debit/credit cards** (e.g., Visa, Mastercard).
- **Mobile wallets** (e.g., bKash, Nagad, Rocket).
- **QR code payments** via apps.

7. What challenges banks are facing in implementing electronic payments in Bangladesh? Describe briefly. 97(4.a)

Challenges Banks Face in Implementing Electronic Payments in Bangladesh

Implementing electronic payments in Bangladesh comes with several challenges that banks need to address. Some of the key challenges are:

1. **Cybersecurity Risks**

- Digital transactions are vulnerable to cyber threats like hacking, phishing, and data breaches.
- Lack of strong cybersecurity measures increases the risk of financial fraud.

2. **Limited Digital Literacy**

- Many users, especially in rural areas, lack awareness and knowledge about digital payment methods.
- Fear of fraud and lack of trust discourage people from using electronic payments.

3. **Infrastructure Limitations**

- Unstable internet connectivity in remote areas affects seamless transactions.
- Lack of widespread ATM and POS (Point of Sale) networks restricts accessibility.

4. **Regulatory and Compliance Issues**

- Banks must follow strict compliance guidelines, which may slow down the adoption of electronic payment systems.
- Frequent policy changes by regulatory authorities create uncertainty.

5. **High Transaction Costs**

- Banks and service providers charge transaction fees, which discourage small businesses and individuals from using electronic payments.
- The cost of maintaining digital payment infrastructure is also high.

While electronic payments offer convenience and efficiency, challenges like security risks, digital illiteracy, and infrastructure gaps need to be addressed. Banks must focus on improving cybersecurity, expanding digital infrastructure, and increasing public awareness to enhance the adoption of electronic payments in Bangladesh.

8. Which payment method, ie, cash, debit/credit cards, MFS, EFT, QR, is the most dominating and which one is the fastest growing in Bangladesh? Explain briefly. 97(5.c)

Most Dominant and Fastest Growing Payment Methods in Bangladesh

Bangladesh's payment ecosystem is evolving rapidly, with **Mobile Financial Services (MFS)** and **digital payment systems** gaining widespread popularity.

Most Dominant Payment Method:

Mobile Financial Services (MFS)

- Platforms like **bKash**, **Nagad**, **Rocket**, and **Upay** are the most widely used for transactions.
- According to Bangladesh Bank, monthly mobile financial transactions exceed **1.5 trillion BDT**.
- MFS is preferred due to its convenience, speed, and security as an alternative to cash transactions.

Fastest Growing Payment Methods:

QR Code & Electronic Fund Transfer (EFT)

- **QR Code payments** are becoming increasingly popular, especially in small businesses and retail transactions.
- **Electronic Fund Transfer (EFT)**, particularly through **BEFTN (Bangladesh Electronic Funds Transfer Network)** and **RTGS (Real-Time Gross Settlement)**, is witnessing rapid adoption.

MFS remains the most dominant payment method in Bangladesh, offering a secure and convenient alternative to cash transactions. Meanwhile, **QR Code payments and EFT are the fastest-growing methods**, contributing to the country's expanding digital payment ecosystem.

9. Distinguish between 'debit card' and 'credit card'. Describe pros and cons of using a cheque as a payment method. 96(3.a)

Distinction between 'Debit Card' and 'Credit Card'

Aspect	Debit Card	Credit Card
Definition	A card linked to the holder's bank account.	A card that allows borrowing money up to a predefined limit.
Fund Source	Funds are deducted from the bank account immediately.	Funds are borrowed from the card issuer and must be repaid.
Payment Process	Payments are processed in real-time.	Payments are made later and interest may be applied.
Spending Limit	Limited to the available balance in the bank account.	Defined by the issuer, can be higher than the available balance.

Interest and Fees	No interest; only withdrawal fees (if any).	Interest is charged on the outstanding balance and late fees may apply.
Usage	Used for everyday purchases, including ATM withdrawals.	Used for larger purchases and credit building.

Pros and Cons of Using a Cheque as a Payment Method

Pros:

1. **Record Keeping:** Provides a paper trail, which is useful for tracking payments and managing finances.
2. **Large Payments:** Convenient for making larger payments, especially when carrying cash is not feasible.
3. **Security:** Can be post-dated or cancelled in case of theft or fraud.
4. **No Transaction Fees:** Unlike credit or debit cards, cheques do not incur transaction fees.

Cons:

1. **Processing Time:** Cheques take longer to clear, usually several days, causing delays in payments.
2. **Risk of Bouncing:** If there are insufficient funds in the account, the cheque will bounce, causing potential fees and penalties.
3. **Requires Bank Account:** Only individuals or businesses with a bank account can issue cheques.
4. **Fraud Risk:** There is a risk of forgery, and they are easier to steal or counterfeit compared to digital payment methods.
5. **Not Widely Accepted:** Many businesses, especially smaller ones, may not accept cheques as a form of payment.

Module-C: Financial System

1. "Discuss the different forms of finance, including Rudimentary, Direct, and Indirect finance. Explain each form with examples."

There are three major forms of finance: **Rudimentary Finance**, **Direct Finance**, and **Indirect Finance**. Each of these methods plays a significant role in how individuals and businesses raise capital or manage financial transactions. Here's a breakdown of each form:

1. Rudimentary Finance

Rudimentary finance refers to basic and simple financial practices that typically involve direct transactions without intermediaries or complex financial instruments. This form of finance is usually informal and does not involve sophisticated financial institutions.

Example:

- A person borrowing money from a friend or family member to meet urgent financial needs.
- Using personal savings or assets to fund a small business without taking loans from banks.

2. Direct Finance

In direct finance, borrowers obtain funds directly from lenders, bypassing any financial intermediaries like banks or financial institutions. Here, the lender provides capital directly to the borrower in exchange for a return on investment (interest or dividends).

Example:

- **Issuing Bonds:** A corporation issues bonds directly to investors in the market, where investors provide funds to the company and, in return, receive periodic interest payments.
- **Stock Issuance:** A company issues shares of stock directly to the public or investors to raise capital for business expansion.

3. Indirect Finance

Indirect finance involves the use of intermediaries such as banks, credit institutions, or other financial organizations that act as a link between borrowers and lenders. The intermediary collects funds from savers (lenders) and provides them to borrowers.

Example:

- A bank collects deposits from customers and then provides loans to individuals or businesses.
- Investment funds that pool money from various investors and lend it out to companies or government bodies.

2. What is a financial system? Which components constitute a financial system? 96(5.a)

Financial system:

A **financial system** is a set of institutions, markets, instruments, and regulations that facilitate the flow of funds between savers and borrowers, helping to allocate resources efficiently within an economy. It plays a crucial role in economic development by channeling savings into investments and facilitating the smooth functioning of financial transactions.

Components of a Financial System:

1. Financial Institutions:

These are intermediaries that help in the transfer of funds between savers and borrowers. They include:

- **Commercial Banks:** Provide services like savings accounts, loans, and credit facilities.
- **Investment Banks:** Help in raising capital, underwriting, and facilitating mergers and acquisitions.
- **Insurance Companies:** Offer policies to protect against financial risks.
- **Pension Funds:** Manage retirement savings and investments.
- **Microfinance Institutions:** Provide small loans to low-income individuals or businesses.

2. Financial Markets:

These are platforms where financial instruments such as stocks, bonds, and other securities are bought and sold. The primary types of markets are:

- **Capital Markets:** Where long-term funding is raised, including stock markets and bond markets.
- **Money Markets:** Where short-term funds are traded, including treasury bills and certificates of deposit.
- **Foreign Exchange Markets:** Where currencies are traded.

3. Financial Instruments:

These are the various types of contracts or assets traded in the financial markets. Some key examples include:

- **Equity Instruments (Shares/Stocks):** Ownership in a company.
- **Debt Instruments (Bonds):** Loans provided to entities, with a promise of repayment with interest.
- **Derivatives:** Financial contracts that derive their value from an underlying asset (e.g., futures, options).

4. Regulatory Bodies:

These are authorities that set and enforce the rules and regulations governing the financial system to ensure its stability and integrity. In Bangladesh, key regulatory bodies include:

- **Bangladesh Bank (BB):** The central bank responsible for monetary policy, regulating banks, and maintaining financial stability.
- **Securities and Exchange Commission (SEC):** Regulates the stock markets and protects investors.
- **Insurance Development and Regulatory Authority (IDRA):** Oversees the insurance sector.

5. Payment Systems:

These are the systems and technologies that facilitate the transfer of money between individuals, businesses, and institutions. Examples include:

- **Electronic Fund Transfer (EFT):** Enables the transfer of funds electronically between banks.
- **Mobile Financial Services (MFS):** Facilitates money transfers through mobile phones, such as bKash, Nagad, etc.

6. Financial Intermediaries:

These are institutions that act as middlemen between savers and borrowers, facilitating the flow of funds. They help channel savings from individuals or businesses to those in need of capital, and in return, they often provide services such as risk management, liquidity, and investment diversification. Examples include:

- **Banks (Commercial and Investment Banks):** They accept deposits from individuals and businesses and provide loans, mortgages, and other financial services.
- **Insurance Companies:** These companies pool funds from policyholders and provide coverage against various risks.

A financial system helps in the efficient allocation of resources, providing liquidity, and enabling businesses and individuals to access capital. It ensures stability in the economy by regulating the movement of funds and protecting the interests of all stakeholders.

3. Differentiate between monetary system, financial system and payment system of a country.

Difference between Monetary System, Financial System, and Payment System of a Country

Aspect	Monetary System	Financial System	Payment System
Definition	The set of institutions, policies and instruments that control the money supply and the value of money in an economy.	The system of financial institutions, markets, instruments, and regulations that facilitate the flow of funds.	The infrastructure and methods that enable the transfer of money between individuals, businesses, and institutions.
Primary Focus	Control and regulation of the money supply and currency.	Mobilization and allocation of financial resources through markets and intermediaries.	Facilitation of the transfer of payments and settlement of transactions.
Key Components	- Central Bank (e.g., Bangladesh Bank) - Money Supply (Currency and deposits) - Interest Rates - Monetary Policies	- Financial Markets (Stock markets, bond markets) - Financial Institutions (Banks, insurance companies, mutual funds) - Financial Instruments (Loans, bonds, stocks) - Regulations	- Electronic Funds Transfer (EFT) - Mobile Payments - Bank Transfers - Payment Cards (Debit/Credit) - QR Code Payments
Role	Ensures price stability, regulates inflation, and maintains economic growth by controlling money supply.	Facilitates savings, investments, and lending. Mobilizes capital and ensures efficient allocation of resources.	Provides the means for making payments and settlements in the economy, ensuring liquidity and efficiency in transactions.
Example	- Central Bank of Bangladesh (Bangladesh Bank) - Currency Circulation - Interest Rate Policies	- Bangladesh Bank, commercial banks, stock markets, mutual funds, insurance companies	- Bank Transfers - Debit/Credit Card Payments - Mobile Financial Services (MFS) - Payment Gateways

4. Describe the financial system of Bangladesh. 99(3.b)

Financial System of Bangladesh

The financial system of Bangladesh is a structure of institutions, markets, instruments, and regulations designed to facilitate the smooth movement of funds between savers and borrowers,

promote investment, ensure economic growth, and stabilize the financial environment. It is a complex and evolving system that has undergone significant reforms over the years.

Components of the Financial System of Bangladesh

1. Financial Institutions:

- **Commercial Banks:** These are the core institutions that provide various financial services like savings and checking accounts, loans, and credit. Some of the major commercial banks include:
 - **State-Owned Banks:** Sonali Bank, Janata Bank, Rupali Bank, etc.
 - **Private Banks:** BRAC Bank, Dutch-Bangla Bank, etc.
 - **Foreign Banks:** Standard Chartered, HSBC, Citibank, etc.
 - **Islamic Banks:** Al-Arafah Islami Bank, Islami Bank Bangladesh Ltd.
- **Non-Banking Financial Institutions (NBFIs):** These include insurance companies, leasing companies, venture capital firms, and mutual funds. Examples are:
 - **Insurance Companies:** Sadharan Bima Corporation, Delta Life Insurance.
 - **Leasing Companies:** Bangladesh Industrial Finance Company Ltd. (BIFC).
- **Microfinance Institutions (MFIs):** MFIs provide financial services to low-income individuals and small businesses that do not have access to traditional banking. Prominent ones include Grameen Bank, BRAC, and ASA.

2. Financial Markets:

- **Money Market:** The money market in Bangladesh deals with short-term funds, primarily for less than one year. It includes instruments like Treasury bills, repurchase agreements, and call money.
- **Capital Market:** The Bangladesh Stock Exchange (DSE and CSE) are the primary stock markets where securities (stocks, bonds) are traded. The **Dhaka Stock Exchange (DSE)** and **Chittagong Stock Exchange (CSE)** are the major bourses. The Bangladesh Securities and Exchange Commission (BSEC) regulates the market.

3. Financial Instruments:

- **Stocks and Bonds:** Equities (stocks) and debt instruments (bonds) are used by businesses to raise capital. The government also issues bonds, such as **Treasury Bonds**, for funding purposes.
- **Debentures:** Companies also issue debentures (long-term debt) to raise funds from the public.

4. Central Bank:

Bangladesh Bank is the central bank and the main regulatory authority for the financial system. It formulates and implements monetary policies, manages foreign exchange

reserves, supervises the commercial banking system, and ensures overall financial stability in the country.

5. Payment Systems:

- Bangladesh has a rapidly evolving payment system that includes traditional methods like **cheques** and **bank drafts** along with modern systems like **mobile financial services (MFS)**, **ATM networks**, **Internet Banking**, **RTGS (Real-Time Gross Settlement)**, and **NPSB (National Payment Switch Bangladesh)**.
- **Mobile Financial Services (MFS)** like **bKash**, **Rocket**, and **Nagad** have become extremely popular, providing financial inclusion for people in remote areas who do not have access to traditional banking.

6. Regulatory Authorities:

- The **Bangladesh Securities and Exchange Commission (BSEC)** regulates and supervises the capital market.
- **The Insurance Development and Regulatory Authority (IDRA)** oversees the insurance industry.
- **The Microcredit Regulatory Authority (MRA)** supervises microfinance institutions.

The financial system of Bangladesh has evolved significantly over the years, with improvements in banking services, capital markets, and payment systems. However, it still faces challenges in terms of regulation, financial inclusion, and infrastructure. With ongoing reforms and the growth of digital financial services, the future of the financial system in Bangladesh looks promising, contributing to the country's economic development.

5. Name the main regulators of the financial system of Bangladesh. Identify the roles of BSEC and BB to develop a vibrant securities market in Bangladesh. 99(6.a)

Main Regulators of the Financial System of Bangladesh

1. **Bangladesh Bank (BB)** – Oversees monetary policy, regulates commercial banks, and ensures financial system stability.
2. **Bangladesh Securities and Exchange Commission (BSEC)** – Regulates the securities market, ensuring transparency and fairness.
3. **Insurance Development and Regulatory Authority (IDRA)** – Regulates the insurance sector.
4. **Microcredit Regulatory Authority (MRA)** – Regulates microfinance institutions.
5. **National Board of Revenue (NBR)** – Regulates tax-related activities in the financial system.

Roles of BSEC and BB to Develop a Vibrant Securities Market

BSEC

1. **Regulates Market Activities** – Ensures transparency and fairness in trading and securities listing.
2. **Protects Investors** – Ensures timely and accurate disclosures, building investor confidence.
3. **Develops Market Infrastructure** – Enhances trading systems, clearing, and settlement processes.

BB

1. **Monetary Policy Management** – Influences interest rates, liquidity, and investment sentiment, impacting the capital market.
2. **Regulates Financial Institutions** – Ensures the stability of institutions participating in the securities market.
3. **Promotes Investor Confidence** – Supports financial institutions to maintain healthy capital levels, fostering market confidence.

Both BSEC and BB contribute significantly to the development of a stable, transparent, and vibrant securities market in Bangladesh.

6. "All forms of money are financial instruments but all financial instruments are not money". Explain. 97(6.a)

"All forms of money are financial instruments, but all financial instruments are not money." Explanation:

Financial Instruments: Financial instruments are contracts that represent a legal agreement between parties for the exchange of money, assets, or services. They include a wide variety of instruments like stocks, bonds, derivatives, loans, and money market instruments. These instruments can be traded or transferred and often serve as a way to raise capital, hedge risk, or transfer ownership.

Money: Money is a specific type of financial instrument that serves as a medium of exchange, a store of value, a unit of account, and a standard of deferred payment. It is universally accepted for transactions and is primarily used for buying goods and services.

Key Differences:

1. **Purpose:**

- **Money** is a financial instrument designed for everyday transactions, enabling the exchange of goods and services.
- **Other financial instruments** (e.g., stocks, bonds) are typically used for investment purposes or to raise capital, not directly for purchasing goods and services.

2. Liquidity:

- **Money** is the most liquid form of financial asset, meaning it can be used immediately for transactions without conversion.
- **Financial instruments**, like stocks or bonds, require conversion (e.g., selling) before they can be used for transactions.

3. Examples:

- **Money:** Currency, coins, central bank reserves.
- **Other financial instruments:** Shares, bonds, futures contracts, options.

While all forms of money are considered financial instruments because they are legally recognized mediums for transactions, not all financial instruments are money. Financial instruments can represent investments, debt, or ownership, whereas money is specifically designed for facilitating exchanges in the economy.

7. What are characteristics of efficient financial market? 98(5.c)

Characteristics of an Efficient Financial Market:

1. Liquidity:
An efficient financial market ensures high liquidity, meaning that assets can be bought and sold quickly without significant price fluctuations. This allows investors to enter and exit positions with ease.
2. Transparency:
The market must have transparent pricing, where all relevant information is publicly available and accessible to all participants. This helps ensure fair pricing and decision-making.
3. Price Discovery:
An efficient market allows for proper price discovery, where the prices of financial assets reflect all available information. This means that asset prices adjust quickly to new information, ensuring that they are fair and represent the true value.
4. Low Transaction Costs:
An efficient financial market minimizes transaction costs, including fees, commissions,

and spreads. This reduces barriers to entry and makes investing more accessible.

5. **Regulatory Oversight and Fairness:**

The market operates under a clear set of rules and regulations that are enforced consistently to prevent manipulation, fraud, and insider trading. This ensures that all participants have equal access and opportunity within the market.

These characteristics ensure that the market functions smoothly, promotes investor confidence, and allows for optimal allocation of resources.

8. Difference between Money market and Capital market.

Difference between Money Market and Capital Market:

Criteria	Money Market	Capital Market
Definition	The money market deals with short-term borrowing and lending of funds, typically with maturities of one year or less.	The capital market involves the buying and selling of long-term financial instruments like stocks and bonds, typically with maturities of more than one year.
Purpose	To manage short-term liquidity and provide a platform for short-term financing.	To provide long-term capital for companies and governments, and facilitate investment opportunities.
Instruments	Treasury bills, certificates of deposit, commercial papers, repurchase agreements.	Stocks, bonds, debentures, and other long-term securities.
Maturity Period	Short-term (up to 1 year).	Long-term (more than 1 year).
Risk Level	Low risk, as instruments are short-term and highly liquid.	Higher risk, as investments are long-term and subject to market fluctuations.
Participants	Commercial banks, central banks, corporations, and financial institutions.	Corporations, governments, investors, and brokers.
Market Type	Primarily wholesale (large sums of money exchanged between financial institutions).	Both wholesale and retail (involves both large institutional investors and individual investors).
Example	Treasury bills, short-term government bonds, commercial papers.	Stocks, corporate bonds, government bonds.

9. Explain financial development in brief. How you can measure the financial development of a country? Illustrate with 3 different methods. 96(5.b)

Financial Development:

Financial development refers to the process by which a country's financial sector expands and improves, enabling the efficient allocation of resources, fostering economic growth, and providing financial services to individuals and businesses. It includes the growth of financial markets, institutions, and instruments, enhancing access to finance and supporting investment and economic activities.

How to Measure the Financial Development of a Country?

The financial development of a country can be measured using various indicators that assess the depth, access, and efficiency of the financial system. Here are three methods to measure it:

1. Financial Depth (Size of the Financial System):

- **Measure:** The ratio of private sector credit to GDP or the ratio of broad money supply (M2) to GDP.
- **Explanation:** A higher ratio indicates a larger and more developed financial sector, showing that a greater proportion of the economy's resources are being allocated through financial markets and institutions.

2. Access to Financial Services:

- **Measure:** The number of bank branches, ATMs, or mobile money accounts per 100,000 people.
- **Explanation:** This indicator reflects how accessible financial services are to the population. More widespread access to banking services indicates better financial inclusion and development.

3. Efficiency of the Financial System:

- **Measure:** The spread between lending and deposit interest rates or the cost of financial transactions.
- **Explanation:** A lower spread or transaction cost suggests a more efficient financial system. Efficiency allows for the better allocation of resources and supports economic growth.

These methods provide a comprehensive view of the financial development in a country by measuring its size, accessibility, and operational efficiency.

10. Give example of five financial development indicators. 99(6.b)

Financial development indicators:

1. **Financial Interrelations Ratio (FIR):** Measures the level of interconnection between different financial sectors (e.g., banks, securities markets). A higher ratio indicates better integration and efficiency within the financial system.
2. **Financial Intermediation and Mobilization Ratio (FIMR):** Shows how efficiently financial institutions mobilize savings and channel them into investments, calculated by the ratio of loans to total deposits. A higher ratio indicates better financial intermediation.
3. **New Issue Ratio (NIR):** Reflects the activity in the capital market by measuring the issuance of new securities. A high ratio indicates a vibrant market for raising capital.
4. **Financialisation Ratio:** Measures the dominance of financial markets and institutions in the economy, indicating a shift towards financial activities over traditional production sectors.
5. **Monetization Ratio:** Compares the money supply (e.g., M2 or M3) to GDP, reflecting how much of the economy is conducted using formal money. A higher ratio signifies a highly monetized economy.

These indicators help assess the development and efficiency of a country's financial system.

11. What are differences between micro credit and micro finance? 98(4.d)

Differences between Microcredit and Microfinance:

Aspect	Microcredit	Microfinance
Definition	Microcredit refers to the provision of small loans to low-income individuals or groups for starting or expanding small businesses.	Microfinance is a broader concept that includes microcredit along with other financial services like savings, insurance, and remittances.
Services Provided	Primarily focused on providing small loans (credit) to individuals.	Provides a range of financial services including loans, savings, insurance, and remittances.
Target Group	Targets individuals or groups with no access to traditional banking, often in rural areas.	Targets low-income individuals and communities, but offers a broader range of services beyond just credit.

Purpose	Primarily to offer access to credit to help individuals start or grow small businesses.	Aims to provide financial inclusion, poverty alleviation, and empower individuals through access to various financial services.
Scope	Narrower in scope, focusing mainly on credit.	Broader in scope, offering a complete package of financial services.

12. Explain the development role of finance.

Development Role of Finance

1. **Development Banking:** Provides long-term financing for infrastructure, industries, and economic growth, supporting national development.
2. **Microcredit:** Offers small loans to low-income individuals, enabling entrepreneurship and poverty reduction.
3. **Microfinance:** Includes credit, savings, and insurance services for financially underserved communities, promoting financial inclusion.
4. **Inclusive Finance:** Ensures financial services reach marginalized groups, enhancing economic participation and reducing inequality.
5. **Sustainable Finance:** Supports environmentally and socially responsible projects, ensuring long-term economic and ecological balance.

13. How transaction costs influence Financial System?

Impact of Transaction Costs on the Financial System

1. **Market Efficiency:** High transaction costs reduce market efficiency by discouraging trading and investment activities.
2. **Liquidity:** Lower transaction costs enhance market liquidity, making it easier to buy and sell financial assets.

3. **Capital Allocation:** High costs can limit access to finance, affecting capital allocation and economic growth.
4. **Financial Inclusion:** Reduced costs promote financial inclusion by making financial services more affordable.
5. **Investment Returns:** Higher transaction costs lower net investment returns, influencing investor behavior and market participation.

14. What are the key functions performed by financial systems? Explain in brief.

Functions of Financial Systems

1. **Mobilization of Savings:** Encourages savings and channels them into productive investments.
2. **Efficient Allocation of Resources:** Ensures capital flows to the most productive sectors.
3. **Risk Management:** Provides financial instruments to manage risks (e.g., insurance, derivatives).
4. **Facilitating Payments:** Supports smooth financial transactions through banking and payment systems.
5. **Providing Liquidity:** Ensures availability of funds for short-term and long-term needs.
6. **Economic Stability:** Regulates financial activities to maintain economic balance and prevent crises.

15. Difference between Spot, Future, Forward and Option markets.

Difference between Spot, Future, Forward, and Option Markets

Market Type	Spot Market	Future Market	Forward Market	Option Market
Definition	Market for immediate purchase and sale of assets.	Contracts to buy/sell an asset at a future date at a predetermined price.	Private agreement to buy/sell an asset at a future date at a specified price.	Contracts giving the right but not the obligation to buy/sell an asset at a future date.
Settlement	Immediate (within two days).	Settled at a future date, with standardization.	Settled at a future date, with customization.	At expiration or earlier; buyer has the option to exercise.

Standardization	Not standardized.	Highly standardized contracts.	Customizable based on needs.	Standardized contracts with strike price and expiry date.
Risk	No risk after the transaction is completed.	Market risk involved until the contract expires.	Counterparty risk due to private agreements.	Limited to the premium paid, but potential for large gains.
Example	Buying stocks today at market price.	Futures contracts for commodities like oil.	Private forward contracts for foreign exchange.	Options on stocks, currencies, or commodities.

16. Difference between Primary and Secondary Market.

Difference between Primary Market and Secondary Market

Criteria	Primary Market	Secondary Market
Definition	Where new securities are issued and sold for the first time.	Where previously issued securities are traded between investors.
Purpose	To raise capital for companies or governments.	To provide liquidity to investors.
Participants	Issuers (companies or governments) and investors.	Investors buying and selling from each other.
Examples	Initial Public Offering (IPO), Bond issuance.	Stock Exchanges (e.g., Dhaka Stock Exchange).
Price Determination	Price is set by the issuer in consultation with underwriters.	Prices are determined by supply and demand in the market.

Module-D: Financial Institutions

1. Explain how Banking Financial Institutions (BFIs) and Non-bank Financial Institutions (NOFIs) serve as financial intermediaries. 96(7.a)

Banking Financial Institutions (BFIs) and Non-bank Financial Institutions (NBFIs) as Financial Intermediaries:

Banking Financial Institutions (BFIs):

1. **Deposits Collection:** BFIs, such as commercial banks, collect deposits from individuals and businesses, creating a pool of funds.
2. **Loans and Credit:** These institutions lend the collected funds to borrowers (individuals, businesses, or governments) for various purposes, like consumption, investment, and development.
3. **Payment Services:** They facilitate the transfer of money through payment systems, offering services like checks, wire transfers, and digital banking.
4. **Risk Management:** Through various financial products like insurance and derivatives, BFIs help manage risks for both depositors and borrowers.
5. **Liquidity Provision:** They ensure liquidity by converting assets into cash quickly through their lending and investment activities.

Non-bank Financial Institutions (NBFIs):

1. **Investment Services:** NBFIs, like insurance companies and mutual funds, pool resources from investors and allocate them into various securities, thus providing capital for businesses and governments.
2. **Credit and Financing:** They offer loans, especially to sectors and individuals that may not have access to traditional banking services, such as microcredit institutions.
3. **Asset Management:** NBFIs, including pension funds and asset management companies, manage portfolios of investments for clients, focusing on long-term growth and returns.
4. **Market Development:** NBFIs provide financial products that complement the banking sector, such as venture capital, leasing, and factoring.
5. **Financial Inclusion:** NBFIs play a key role in promoting financial inclusion by providing access to financial services for underserved populations.

2. Outline the functions and growth of BFIs in Bangladesh. 96(7.b)

Functions and Growth of Banking Financial Institutions (BFIs) in Bangladesh:

Functions of BFIs in Bangladesh:

1. **Mobilization of Savings:** BFIs collect savings from individuals, businesses, and government entities in the form of deposits, which are then used for lending and investment purposes.
2. **Credit Provision:** They provide loans and advances to various sectors such as agriculture, industry, and services, fostering economic growth.
3. **Payment Services:** BFIs facilitate various payment mechanisms, including electronic transfers, checks, and remittances, ensuring smooth financial transactions.
4. **Risk Management:** They offer products like insurance and hedging services to manage financial risks and protect assets.
5. **Investment Services:** BFIs invest in government bonds, securities, and other financial instruments to generate returns and diversify their portfolios.

Growth of BFIs in Bangladesh:

1. **Expansion of Banking Network:** The number of commercial banks and branches has increased significantly, providing access to banking services in both urban and rural areas.
2. **Adoption of Technology:** The introduction of digital banking and mobile financial services has helped in financial inclusion, enabling remote areas to access banking services.
3. **Improved Financial Products:** BFIs have introduced a wide range of financial products like Islamic banking, mobile banking, and microfinance services to cater to diverse customer needs.
4. **Strengthening Capital Base:** With regulatory measures like capital adequacy ratios and strict supervision, BFIs in Bangladesh have strengthened their financial stability.
5. **Increased Foreign Investment:** The entry of foreign banks and financial institutions has brought in capital, expertise, and technology, contributing to the overall growth of BFIs.

The growth and expansion of BFIs in Bangladesh have been pivotal in supporting the country's economic development and financial inclusion.

3. Outline the functions and growth of NBFIs in Bangladesh. 96(7.b)

Functions and Growth of NBFIs in Bangladesh:

Functions of NBFIs:

1. **Credit Provision:** NBFIs offer loans and credit facilities to individuals, businesses, and industries, particularly focusing on small and medium-sized enterprises (SMEs) that may not have access to traditional bank financing.
2. **Investment Services:** They provide investment options in bonds, securities, and stocks, offering diverse opportunities for capital growth.
3. **Insurance Services:** NBFIs offer life and non-life insurance products, contributing to risk management for individuals and businesses.
4. **Microfinance:** They extend microcredit and microfinance services to low-income and underserved segments, promoting financial inclusion.
5. **Asset Management:** NBFIs manage mutual funds and pension funds, helping investors achieve long-term financial goals.

Growth of NBFIs in Bangladesh:

1. **Increased Market Share:** The sector has seen a significant increase in market participation, particularly in providing loans and microfinance services to underprivileged populations.
2. **Technological Advancements:** NBFIs have adopted modern technologies like mobile banking and digital platforms to expand their services and improve customer access.
3. **Regulatory Support:** Government regulations and the central bank's policy support have facilitated the growth and stability of NBFIs.
4. **Rising Investment Products:** There has been a growth in investment products such as mutual funds and bond markets, increasing investor choices.
5. **Enhanced Public Awareness:** The increasing public awareness of NBFIs services has led to their growing popularity in the financial system.

NBFIs play a crucial role in enhancing financial inclusion, fostering economic growth, and supporting underdeveloped sectors in Bangladesh.

4. What are the different roles banks play in an economy?

Roles Banks Play in an Economy:

1. **Facilitating Payments:** Banks provide payment services, making transactions easier for individuals and businesses, such as through checks, credit cards, and electronic transfers.
2. **Mobilizing Savings:** Banks encourage saving by offering interest on deposits, which helps channel funds into productive investments.
3. **Providing Credit:** Banks lend money to individuals and businesses, promoting investment and consumption, which drives economic growth.
4. **Managing Risks:** Banks offer financial products like insurance and derivatives to help individuals and businesses manage risks.
5. **Monetary Policy Implementation:** Banks play a key role in implementing a country's monetary policy by controlling the money supply, influencing interest rates, and supporting government objectives like inflation control and economic stability.

5. What are the different aspects of Asset-Liability Management (ALM)?

Different Aspects of Asset-Liability Management (ALM):

1. **Liquidity Management:** Ensures that a financial institution has enough liquid assets to meet its short-term obligations.
2. **Interest Rate Risk Management:** Focuses on managing the impact of interest rate fluctuations on the balance sheet by matching assets and liabilities effectively.
3. **Capital Adequacy:** Ensures that a financial institution maintains sufficient capital to cover potential losses and meet regulatory requirements.
4. **Profitability Management:** Strives to optimize the return on assets while managing risks to maintain profitability in both short and long terms.
5. **Regulatory Compliance:** Ensures that the institution complies with regulations related to asset-liability matching and risk management.

These aspects together help financial institutions manage their risks, maintain stability, and ensure long-term financial health.

6. What are the key trends in commercial banking?

Key Trends in Commercial Banking:

1. **Digital Transformation:** Commercial banks are increasingly adopting digital technologies like mobile banking, online banking, and AI-driven solutions to improve customer experience and operational efficiency.
2. **Fintech Integration:** Collaboration with financial technology firms is rising, enabling banks to offer innovative products and services, such as peer-to-peer lending and digital wallets.
3. **Customer-Centric Services:** Banks are focusing on personalized services to meet the diverse needs of customers, using data analytics to tailor their offerings.
4. **Regulatory Compliance:** Banks are placing greater emphasis on adhering to stricter regulatory requirements and ensuring transparency in operations.
5. **Sustainable Banking Practices:** Increasing attention is being given to environmentally and socially responsible investments, with many banks incorporating ESG (Environmental, Social, and Governance) criteria into their portfolios.

Module-E: Financial Markets

1. What is the purpose of the money markets?

The purpose of the money markets:

The purpose of the money markets is to facilitate short-term borrowing and lending of funds with high liquidity and low risk. These markets serve several key functions:

1. **Liquidity Management:** Money markets help financial institutions, businesses, and governments manage their short-term liquidity needs by providing a platform for borrowing and lending funds.
2. **Efficient Fund Allocation:** They allocate funds to entities with temporary cash shortfalls while providing surplus funds to those who need them.
3. **Interest Rate Determination:** Money markets play a significant role in determining short-term interest rates, which influences broader economic conditions.
4. **Short-Term Financing:** These markets offer instruments like Treasury bills and repurchase agreements (repos) for businesses and governments to meet their short-term funding requirements.
5. **Risk Reduction:** Money markets are low-risk markets where investors can park their funds for short durations without significant exposure to risk.

In summary, money markets ensure liquidity, help manage short-term funding needs, and contribute to the overall stability of the financial system.

2. Who are the participants in the money markets and what are their roles?

Participants in the money markets and their roles:

1. Central Banks:

Role: They regulate money supply and control short-term interest rates to stabilize the economy. Central banks also intervene to maintain liquidity in the system.

2. Commercial Banks:

Role: They participate by borrowing and lending short-term funds to meet their liquidity needs. They also act as intermediaries in the money market.

3. Corporations:

Role: Large corporations borrow or invest in short-term instruments like commercial papers to manage their short-term funding needs.

4. Money Market Mutual Funds:

Role: They pool funds from individual investors and invest in short-term debt securities, providing liquidity and safety for their investors.

5. Governments:

Role: Governments issue short-term securities such as Treasury bills to manage their short-term funding needs, while also serving as a low-risk investment option for other participants.

Each participant plays a key role in ensuring liquidity, efficiency, and stability within the money markets.

3. What are the key instruments used in the money markets?

The key instruments used in the money markets are:

1. Treasury Bills (T-Bills): Short-term debt securities issued by the government to raise funds, typically with maturities of 91, 182, or 364 days.
2. Certificates of Deposit (CDs): Time deposits offered by banks with fixed interest rates and maturity dates, usually ranging from a few weeks to a year.
3. Commercial Paper: Unsecured, short-term debt issued by corporations to meet their working capital needs, typically maturing in 30 to 270 days.
4. Repurchase Agreements (Repos): Short-term borrowing agreements where securities are sold with an agreement to repurchase them at a later date, often used by financial institutions.
5. Banker's Acceptances: Short-term debt instruments issued by a bank on behalf of a borrower, typically used in international trade, guaranteeing payment.

These instruments help in managing short-term funding and liquidity in the money markets.

4. What is the capital market and what are its functions?

Capital Market:

Capital Market is a part of the financial market where long-term debt or equity-backed securities are bought and sold. It provides companies, governments, and other organizations with access to long-term financing by issuing stocks, bonds, and other securities.

Functions of the Capital Market:

1. **Mobilization of Savings:** It channels savings into productive investments by offering investment opportunities.
2. **Facilitating Investment:** It helps businesses raise capital for expansion, research, and development by issuing securities.
3. **Price Discovery:** It ensures proper pricing of securities based on supply and demand in the market.
4. **Liquidity:** It provides liquidity to investors by allowing them to buy and sell securities easily.
5. **Economic Growth:** It contributes to the overall economic growth by fostering investment and efficient capital allocation.

5. Who are the participants in the capital market?

Participants in the Capital Market:

1. **Issuers:** Entities such as companies, governments, and other organizations that issue securities (stocks, bonds) to raise capital.
2. **Investors:** Individuals or institutions who purchase securities in the capital market, seeking returns on their investments.
3. **Brokers:** Intermediaries who facilitate the buying and selling of securities between investors and issuers.
4. **Dealers:** Financial institutions or individuals who buy and sell securities for their own account, providing liquidity to the market.
5. **Regulatory Authorities:** Government agencies, such as the Securities and Exchange Commission (SEC), that oversee and regulate the capital markets to ensure fairness and transparency.

6. What is the bond market and how does it operate?

Bond Market:

The **bond market** is a financial market where participants buy and sell debt securities, primarily bonds. Bonds are issued by corporations, governments, or other entities as a way to raise capital. In return, the issuer agrees to pay periodic interest (coupon) payments and to repay the principal amount (face value) at maturity.

How It Operates:

1. **Issuance:** Bonds are issued by corporations, governments, or municipalities to raise funds for various purposes like infrastructure development, expansion, or funding projects.
2. **Trading:** Bonds can be traded in both primary and secondary markets. In the primary market, investors purchase bonds directly from issuers. In the secondary market, investors buy and sell bonds among themselves.
3. **Pricing:** Bond prices are influenced by factors like interest rates, credit ratings, and economic conditions. As interest rates rise, bond prices typically fall, and vice versa.
4. **Yield:** The bond's yield is the return an investor can expect based on the bond's price and coupon rate. It is an essential factor in determining bond investment attractiveness.
5. **Maturity:** Bonds have a set maturity period, at the end of which the principal amount is paid back to the bondholder.

7. What is the distinction between interest rates and returns in the financial markets?

Distinction between Interest Rates and Returns in Financial Markets:

Aspect	Interest Rate	Return
Definition	The percentage charged on borrowed money or paid on deposits.	The gain or loss on an investment over a period, expressed as a percentage.
Focus	Cost of borrowing or income from lending.	Overall profitability or performance of an investment.
Scope	Fixed or variable, based on lending agreements or central bank policies.	Can fluctuate based on asset performance, market conditions, and type of investment.
Nature	Typically a simple percentage, often annual, reflecting cost or income for a set period.	Broader measure including interest, dividends, or capital appreciation.
Calculation	Based on the principal amount, usually predetermined.	Based on changes in investment value, including income, dividends, and capital gains.

Module-F: Islamic Financial System

1. What are the Islamic governing principles (Shariah) in finance?

Islamic Governing Principles (Shari'ah) in Finance

1. **Prohibition of Riba (Interest)** – Any form of interest or usury is strictly forbidden in Islam, promoting profit-sharing instead of debt-based financing.
2. **Avoidance of Gharar (Uncertainty)** – Contracts with excessive uncertainty, speculation, or deception are not allowed to ensure fairness in transactions.
3. **Prohibition of Maysir (Gambling)** – Engaging in speculative activities or games of chance is prohibited as it creates unjust enrichment.
4. **Zakat (Wealth Distribution)** – A mandatory charitable contribution is required to promote social justice and economic equality.
5. **Ethical Investments** – Islamic finance encourages investments in socially responsible and halal (permissible) businesses, avoiding industries like alcohol, gambling, and pork-related products.

2. What do you know about Shari'ah governance in an Islamic financial institution? Outline the key players and their role in a 'Shari'ah governance framework'. 96(9.c)

Shari'ah Governance in an Islamic Financial Institution

Shari'ah governance ensures that Islamic financial institutions (IFIs) comply with Islamic principles. It establishes a framework for oversight, accountability, and decision-making in line with Shari'ah laws.

Key Players and Their Roles

1. **Shari'ah Board** – Composed of Islamic scholars who provide guidance on Shari'ah compliance, issue fatwas, and oversee financial products.
2. **Board of Directors** – Responsible for overall governance, ensuring that the institution follows Shari'ah principles in its operations.
3. **Management** – Implements Shari'ah-compliant policies and ensures daily operations align with Islamic finance principles.

4. **Shari'ah Compliance Department** – Monitors and reviews financial activities, ensuring adherence to Shari'ah guidelines and reporting to the Shari'ah Board.
5. **Shari'ah Audit Function** – Conducts independent audits to verify compliance with Shari'ah rulings and identifies any non-compliant activities.
6. **Regulatory Authorities** – Set legal and regulatory frameworks for Shari'ah governance, ensuring IFIs operate within Islamic financial laws.

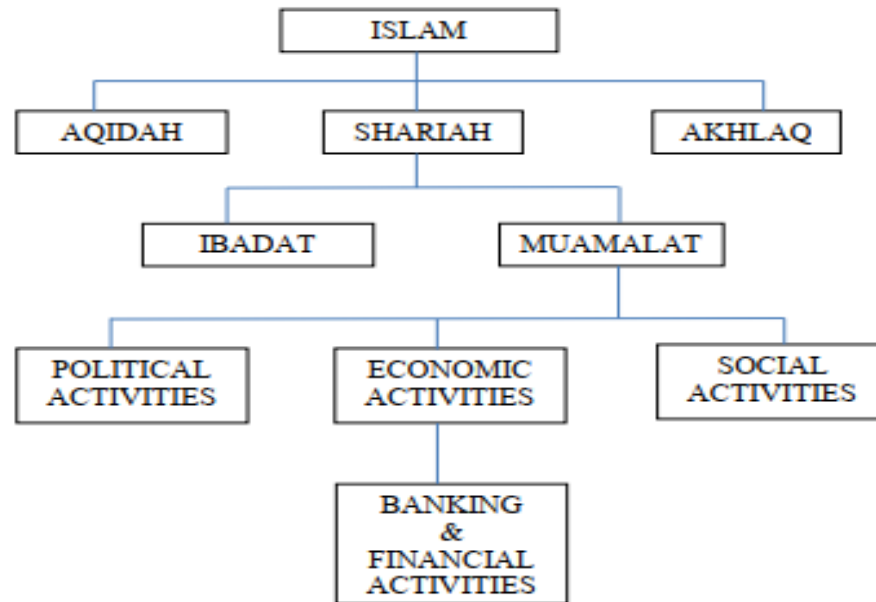
3. What is the relationship between Finance and Religion (Islam)?

Relationship between Finance and Religion (Islam)

Islamic finance is deeply connected to religion through **Shari'ah principles**, ensuring ethical and fair financial practices. The relationship can be explained as follows:

1. **Three Core Elements**: Islamic finance is based on **Aqidah (faith)**, **Shari'ah (laws)**, and **Akhlaq (ethics)**, ensuring financial dealings align with Islamic values.
2. **Muamalat (Transactions)**: Islamic finance falls under **Muamalat**, which governs economic, social, and political activities, ensuring fairness in financial transactions.
3. **Prohibition of Riba and Gharar**: Islam forbids **riba (interest)** and **gharar (excessive uncertainty)** to promote risk-sharing and transparency in finance.
4. **Zakat and Social Welfare**: Finance is linked to **Zakat**, ensuring wealth distribution to support economic justice and reduce inequality.
5. **Ethical Investments**: Islamic finance prohibits investments in **haram (forbidden)** activities such as gambling, alcohol, and speculation, ensuring socially responsible financial growth.
6. **Profit and Risk Sharing**: Instead of interest-based transactions, Islamic finance promotes equity-based partnerships like **Mudarabah (profit-sharing)** and **Musharakah (joint ventures)**, aligning finance with ethical and religious principles.

The Relationship Between the Banking System and Religion



4. Briefly explain the basic fundamentals of Shari'ah Law? List major principles that must be followed while developing Shari'ah compliant products. 96(9.a)

Basic Fundamentals of Shari'ah Law

Shari'ah law is a comprehensive system of guidance that governs all aspects of a Muslim's life, including ethics, social conduct, and legal matters. It is derived from the Qur'an, Hadith, Ijma, and Qiyas. The basic fundamentals include:

1. **Tawhid (Oneness of God)**: Everything in a Muslim's life, including financial dealings, should align with the belief in the oneness of Allah.
2. **Adherence to Qur'an and Hadith**: The Qur'an is the eternal and primary source of Shari'ah law, and the Hadith provides the teachings and practices of Prophet Muhammad.
3. **Ijma (Consensus)**: The agreement of scholars on matters of law is an essential source when Qur'an and Hadith do not provide a clear answer.
4. **Qiyas (Analogical Reasoning)**: Scholars apply reasoning by analogy to solve new issues based on precedents set by the Qur'an and Hadith.
5. **Ethics and Morality**: Shari'ah emphasizes the importance of ethical conduct, justice, and welfare for society.

Major Principles for Developing Shari'ah-Compliant Products

1. **No Riba (Interest):** All financial products must avoid interest-based transactions, promoting profit-sharing or risk-sharing arrangements.
2. **Ethical and Halal Practices:** Products must not involve prohibited activities such as alcohol, gambling, or unethical business practices.
3. **Risk Sharing:** Financial transactions should share risk between parties, rather than transferring the entire risk to one side.
4. **Transparency:** Clear and straightforward terms must be stated in all contracts to avoid Gharar (excessive uncertainty).
5. **Public Interest (Maslahat):** Products should serve the broader good and contribute to the welfare of society.

These principles ensure that financial products comply with Shari'ah law, focusing on justice, fairness, and ethical conduct.

5. How come Ijma & Qiyas (as source of Shariah law) are different from Quran and Hadith?
98(9.d)

How Ijma & Qiyas are Different from Quran and Hadith

1. **Ijma:** Ijma refers to the consensus or unanimous agreement of Islamic scholars (mujtahids) on a specific legal issue not explicitly addressed by the Qur'an or Hadith. Unlike the Qur'an and Hadith, which provide direct and clear guidelines, Ijma is based on scholarly reasoning and collective judgment on matters that were not directly mentioned in the foundational texts.
2. **Qiyas:** Qiyas is the process of analogical reasoning, where scholars compare an issue or situation not covered by the Qur'an and Hadith to one that has already been addressed in these primary sources. Qiyas is essentially the application of reasoning to deduce a solution by drawing parallels to known cases. Unlike the Qur'an and Hadith, which are direct sources of law, Qiyas involves human judgment and is used to address novel issues that arise in society.
3. **Quran:** The Qur'an is the divine and eternal source of Shariah, containing explicit laws and teachings revealed by Allah to the Prophet Muhammad (PBUH). It is considered the ultimate and most authoritative source in Islamic law, and its guidance is infallible.
4. **Hadith:** The Hadith consists of the sayings, actions, and approvals of the Prophet Muhammad (PBUH). Like the Qur'an, it serves as a primary source of guidance but is secondary to the Qur'an in authority. Hadith provides explanations and clarifications of the Qur'an's principles and offers practical examples for implementation.

Role of Money in Islamic Economics and Finance

In Islamic economics and finance, money is viewed primarily as a **means of exchange** and a **unit of value**, rather than as a commodity or object of profit. Several key principles define the role of money in this context:

1. **Means of Exchange**: Money facilitates trade by overcoming the limitations of a barter system. It allows for easier transactions and the exchange of goods and services, promoting economic activity and growth.
2. **Measure of Value**: Money is used to measure the value of goods and services, providing a standard unit that helps in evaluating and comparing different products in the economy.
3. **Money Should Not Be Hoarded**: Money must not be hoarded or withdrawn from circulation. This ensures that it remains active in facilitating economic exchanges and prevents negative economic consequences such as deflation or market stagnation.
4. **Prohibition of Riba (Interest)**: Islamic finance strictly forbids the practice of earning money through interest (riba). Money itself does not have intrinsic value, and profit should only be earned through legitimate trade, investment in tangible assets, or business ventures, not from lending money or charging interest.
5. **Equity-Based Finance**: Instead of debt-based financial systems, Islamic finance emphasizes equity-based structures where profit and loss are shared between investors and entrepreneurs. This promotes fairness and reduces the risk of exploitation and unethical wealth accumulation.
6. **Backed by Tangible Assets**: Islamic economics encourages investment in real economic activities, where financial transactions are tied to tangible assets, such as property or business ventures. This contrasts with conventional finance, where transactions may not be linked to real assets and can lead to speculative activities and financial crises.

In essence, money in Islamic finance serves as a tool for facilitating exchange and measuring value, but it is not an object for generating profit through interest. It must be used productively, tied to real economic activities, and involve equity-based structures to promote fairness and reduce financial instability.

7. Distinguishing the features of Islamic Banking.

Distinguishing Features of Islamic Banking

Islamic banking has several unique characteristics that differentiate it from conventional banking. According to Chapra, the following are five key distinguishing features:

1. **Abolition of Interest (Riba):** Islamic banks are interest-free, as riba (interest) is prohibited in the Holy Quran. All forms of interest are avoided in Islamic banking.
2. **Adherence to Public Interest:** Islamic banks serve the public interest, focusing on socio-economic goals rather than individual or group profits. Deposits from the public are used to benefit society as a whole.
3. **Multi-purpose Bank:** Islamic banks are universal and offer a wide range of services, combining elements of commercial and investment banking. They finance specific projects or ventures and do not rely on short-term borrowing.
4. **More Careful Evaluation of Investment Demand:** Islamic banks place great emphasis on evaluating equity-oriented investments. Unlike conventional banks, they assess risks and financing requests carefully, promoting risk-sharing.
5. **Catalyst of Development:** Islamic banks foster closer relationships with entrepreneurs through profit-loss sharing. They provide technical consultation and financial advice, contributing to industrialization and development.

8. Briefly explain the objectives of Islamic Banking. How Islam interprets 'Riba'?

Objectives of Islamic Banking

Islamic banking aims to promote financial services in line with Islamic principles and contribute to economic development. The key objectives are:

1. **Offer Shari'ah-compliant Financial Services:** Islamic banking provides financial transactions that comply with Islamic law, avoiding interest (riba) and ensuring ethical practices.
2. **Contribute to Economic Development:** Islamic banking supports development by linking profit-sharing with successful business outcomes, promoting entrepreneurship and growth.
3. **Optimum Resource Allocation:** Islamic banks aim to allocate financial resources efficiently into projects that benefit the economy, focusing on profitability and social benefit.

4. **Equitable Distribution of Income:** It seeks to ensure fair income distribution between the bank, depositors, and entrepreneurs, avoiding exploitation and promoting social justice.

Islam's Interpretation of 'Riba'

In Islam, **riba** refers to the practice of charging or receiving interest on loans, which is considered exploitative and unjust. The charging of interest is prohibited as it leads to unfair profit accumulation without any real value exchange, contradicting the principles of fairness and equity in Islam.

9. What are the key principles of Islamic banking? What are the conceptual differences of 'interest' and 'riba'? 97(7.a)

Key Principles of Islamic Banking

1. **Prohibition of Riba (Interest):** Islamic banking prohibits earning or paying interest on loans, as it is considered exploitative.
2. **Profit and Loss Sharing:** Transactions are based on equity and risk-sharing models such as Mudarabah (profit-sharing) and Musharakah (joint venture), where both the bank and the customer share the profits or losses.
3. **Asset-Backed Financing:** Transactions in Islamic banking must be supported by tangible assets or services. Speculation (Gharar) is prohibited.
4. **Ethical Investments:** Islamic banks only invest in businesses or projects that comply with Islamic principles (e.g., no alcohol, gambling, or unethical practices).
5. **Risk Sharing:** Both the bank and the customer share the risks and rewards of investments, promoting fairness and mutual benefit.

Conceptual Differences between 'Interest' and 'Riba'

- **Interest:** It is the amount paid by a borrower to a lender for the use of money, typically calculated as a percentage of the loan amount. It is considered exploitative in conventional banking.
- **Riba:** In Islam, Riba is broader than just interest; it refers to any guaranteed return on money without taking any risk. Riba includes both interest and any excessive increase in money that doesn't result from actual trading or investment, making it unjust.

10. Based on principles, what are the available Islamic financial instruments? How do you differentiate between them? 98(9.c)

Islamic Financial Instruments:

1. **Mudaraba:** A profit-sharing agreement between an investor and an entrepreneur. The investor provides capital, while the entrepreneur manages the business. Profits are shared based on an agreed ratio, while losses are borne by the investor.
2. **Musharaka:** A partnership arrangement where both parties contribute capital and share the profits and losses according to their respective shares. It may involve joint management of the business.
3. **Murabaha:** A cost-plus financing arrangement where the bank purchases goods on behalf of the customer and sells them at a marked-up price. The price and profit margin are agreed upon upfront.
4. **Bai Mua'zzal:** A variant of Murabaha, where the borrower can defer payment for the goods purchased within a specific period, with mutually agreed terms.
5. **Ijara:** A lease agreement where the bank buys an asset and leases it to the customer for an agreed rental amount and period. The ownership remains with the bank.
6. **Ijara wa-iktina:** A lease-purchase agreement where the bank leases an asset to the customer with an agreement to transfer ownership after the rental payments.
7. **Qard Hassan:** A benevolent loan where the borrower repays only the principal amount. The borrower may give additional rewards, but it is not mandatory.
8. **Wadiah:** A trust arrangement where assets are deposited with a custodian who is allowed to use the assets under agreed conditions.
9. **Rahn:** A pledge or pawn where an asset is pledged as security for a loan. Ownership remains with the borrower, but the pledgee has the right to take possession under certain conditions.

Differentiation between These Instruments:

- **Profit and Loss Sharing:** Mudaraba and Musharaka focus on shared profits and losses, whereas the other instruments typically involve fixed returns or specific terms.
- **Fees or Charges Based:** Murabaha, Bai Mua'zzal, Ijara, and Ijara wa-iktina involve charges or fees like cost-plus pricing, leasing, and deferred payments.
- **Benevolent and Trust-Based:** Qard Hassan is a benevolent loan with no profit, while Wadiah and Rahn focus on the safekeeping and pledge of assets without transferring ownership.

11. List three key Shariah Principles to adopted by Islamic banks in delivering their products.

Three Key Shariah Principles Adopted by Islamic Banks:

1. **Mudaraba:** Profit-sharing or trust finance where one party provides capital, and the other manages the business, sharing the profit based on a pre-agreed ratio.
2. **Musharaka:** Partnership or joint venture where both parties contribute capital, and profits and losses are shared based on their investment ratio.
3. **Murabaha:** Cost-plus financing, where the bank buys goods and resells them to the customer at a marked-up price, with the profit margin disclosed to the customer.

12. What are the investment products of Islamic banks? Explain the key features of the investment products of Islamic banks in Bangladesh State and explain the fundamental sources of Shariah Law. 97(7.b)

Investment Products of Islamic Banks in Bangladesh:

1. **Mudaraba Investment Accounts:** This is a profit-sharing agreement where the bank acts as the investor (Rabbul-Mal) and the account holder acts as the entrepreneur (Mudarib). The profits generated from the investment are shared based on an agreed-upon ratio, but in case of loss, the bank bears the loss (except for negligence).
2. **Musharaka Investment Accounts:** This is a joint venture between the bank and the customer. Both parties contribute capital and share the profits and losses based on the agreed ratio. It encourages partnership and collaboration.
3. **Murabaha Financing:** The bank purchases goods on behalf of the customer and sells them at a higher price, where the price includes a profit margin. The customer pays the agreed price in installments.
4. **Ijara and Ijara wa-iktina (Lease Financing):** Ijara is lease-based financing where the bank buys an asset and leases it to the customer for a fixed period. In Ijara wa-iktina, there is an option for the customer to purchase the leased asset at the end of the lease term.
5. **Islamic Bonds (Sukuk):** Sukuk are asset-backed securities that provide returns based on profits from underlying assets, as opposed to traditional bonds, which pay interest.

Key Features of Investment Products in Bangladesh:

- **Shariah Compliance:** All products must comply with Islamic principles, such as the prohibition of interest (riba) and uncertainty (gharar).

- **Profit-sharing:** Most products are based on the concept of profit-sharing, emphasizing risk-sharing and mutual benefit.
- **Asset-backed Financing:** Investment products often involve tangible assets, ensuring that all financing is backed by physical assets.

Fundamental Sources of Shariah Law:

1. **The Quran:** The primary source, providing divine guidance and the basis for Islamic law.
2. **Hadith (Sunnah):** The sayings and actions of Prophet Muhammad (PBUH), which provide detailed interpretations of the Quran.
3. **Ijma (Consensus):** The consensus of Islamic scholars on a particular issue when there is no explicit ruling in the Quran or Hadith.
4. **Qiyas (Analogical Reasoning):** The use of analogy to derive rulings on issues not directly addressed in the Quran or Hadith.

13. What are the major prohibitions that must be followed while developing a shariah compliant financial services model? How sukuk is different from commercial bonds?

Major Prohibitions in Shariah-Compliant Financial Services Model:

1. **Riba (Interest):**
Any form of interest (riba) is strictly prohibited. Financial transactions that involve guaranteed returns on capital or interest-based income are not allowed.
2. **Gharar (Uncertainty):**
Transactions with excessive uncertainty or ambiguity (gharar) are prohibited. This includes contracts where the terms or the subject matter are unclear, leading to risk.
3. **Maysir (Gambling):**
Speculative activities, such as gambling, are forbidden. This includes trading in options, derivatives, or any other financial product with a speculative nature.
4. **Investment in Haram (Forbidden) Industries:**
Investment in industries or businesses related to alcohol, gambling, pork, and other haram (forbidden) sectors is not permissible.
5. **Unlawful Contracts:**
Contracts based on illegal terms or unethical practices, such as selling goods without ownership, are not allowed.

Differenc between Sukuk and Commercial Bonds:

Feature	Sukuk	Commercial Bonds
Nature of the Instrument	Asset-backed securities, representing ownership in tangible assets or business ventures.	Debt instruments, representing a loan with a promise to repay principal with interest.
Shariah Compliance	Fully compliant with Shariah; no riba (interest), and linked to lawful economic activities.	Not Shariah-compliant; involves riba (interest).
Return Mechanism	Return based on profit from underlying assets or business ventures.	Return is fixed interest paid over time.
Ownership Rights	Sukuk holders have partial ownership in the underlying asset/project.	Bondholders are creditors and do not have ownership rights.
Risk and Reward	Sukuk holders share in the risk and reward of the underlying asset.	Bondholders do not share in the project's profits; they only receive fixed interest.

Module-G: Regulatory Framework for Financial, Monetary and Payment System

1. Why is financial system regulation necessary?

Why is Financial System Regulation Necessary:

1. **Ensures Stability:** Regulation helps maintain stability within the financial system by preventing risks that can lead to financial crises.
2. **Protects Consumers:** It ensures that financial products and services are safe for consumers, preventing fraud and exploitation.
3. **Promotes Fairness:** Regulation ensures that all financial institutions operate on a level playing field, preventing unethical practices.
4. **Encourages Confidence:** A well-regulated system builds trust in the financial sector, encouraging investment and economic growth.

2. What are the different types of financial system regulations?

Types of Financial System Regulations:

1. **Microprudential Regulation:**

- Focuses on the stability and risk management of individual financial institutions.
- Ensures banks and financial firms maintain adequate capital and liquidity to prevent failures.
- Example: Basel III capital adequacy requirements.

2. **Macroprudential Regulation:**

- Aims to protect the financial system as a whole by addressing systemic risks.
- Focuses on preventing financial crises by monitoring market-wide risks and interconnections.
- Example: Counter-cyclical capital buffers to control excessive credit growth.

3. What are the key objectives of the central bank?

Key Objectives of the Central Bank:

1. **Price Stability:**

- Controlling inflation and deflation to maintain a stable price level in the economy.
- Ensuring purchasing power stability to support economic activities.

2. **Full Employment:**

- Promoting policies that support job creation and reduce unemployment.
- Maintaining a balance between economic growth and labor market stability.

3. **Financial Stability:**

- Ensuring the stability of banks and financial institutions to prevent financial crises.
- Supervising and regulating the banking sector to reduce systemic risks.

4. **Economic Growth:**

- Implementing monetary policies to support sustainable economic growth.
- Facilitating credit availability to businesses and industries.

5. **Exchange Rate Stability:**

- Managing foreign exchange reserves and interventions to stabilize the currency.
- Preventing excessive volatility in exchange rates to promote trade and investment.

4. What are the main functions of the central bank?

Main Functions of the Central Bank:

1. **Monetary Management:**

- Implements monetary policy to control inflation and money supply.
- Regulates interest rates to stabilize the economy.

2. **Issuance of Currency:**

- Has the sole authority to issue national currency.
- Ensures adequate money supply to meet economic needs.

3. **Banking Supervision:**

- Regulates and supervises banks to maintain financial stability.
- Ensures banks follow prudential regulations and risk management practices.

4. **Lender of Last Resort:**

- Provides emergency funding to banks facing liquidity crises.
- Prevents financial system collapse during economic shocks.

5. **Foreign Exchange and Reserves Management:**

- Manages foreign exchange reserves and exchange rate stability.
- Implements policies to facilitate international trade and investment.

6. **Developmental Role:**

- Supports financial inclusion and promotes economic development.
- Encourages innovation and digital banking initiatives.

5. What are the methods of credit control used by the central bank?

Methods of Credit Control Used by the Central Bank:

The central bank uses two main types of credit control methods to regulate money supply and credit availability in the economy:

1. Quantitative Credit Control Methods: (Regulate overall credit volume)

- Bank Rate Policy:** Adjusting the interest rate at which the central bank lends to commercial banks.
- Open Market Operations (OMO):** Buying and selling government securities to control liquidity.
- Cash Reserve Ratio (CRR):** Mandating banks to maintain a percentage of their deposits with the central bank.
- Statutory Liquidity Ratio (SLR):** Requiring banks to keep a portion of their funds in cash, gold, or approved securities.

2. Qualitative Credit Control Methods: (Regulate credit direction and usage)

- Selective Credit Control:** Restricting credit for speculative purposes like stock market investments.

- ii. **Moral Suasion:** Persuading banks to follow credit policies for economic stability.
- iii. **Credit Rationing:** Limiting the amount of credit banks can lend to specific sectors.

These methods help in stabilizing inflation, ensuring economic growth, and maintaining financial stability.

6. Who are the regulators of the financial system in Bangladesh?

Regulators of the Financial System in Bangladesh:

1. **Bangladesh Bank (BB):** Regulates and supervises banks, monetary policy, and foreign exchange.
2. **Bangladesh Securities and Exchange Commission (BSEC):** Oversees the capital market, stock exchanges, and securities regulations.
3. **Insurance Development and Regulatory Authority (IDRA):** Regulates the insurance sector, ensuring policyholder protection.
4. **Microcredit Regulatory Authority (MRA):** Supervises and regulates microfinance institutions (MFIs).
5. **Financial Reporting Council (FRC):** Ensures transparency and accountability in financial reporting and auditing.

7. What are the core policies of the central bank?

Core Policies of the Central Bank:

1. **Monetary Policy:**
It involves managing the money supply, interest rates, and inflation to maintain price stability and control inflation.
2. **Exchange Rate Policy:**
The central bank manages the currency exchange rate to ensure stability in foreign trade and protect the currency's value.
3. **Financial Stability Policy:**
This aims to ensure the stability of the financial system by monitoring risks, regulating financial institutions, and maintaining liquidity.
4. **Banking Supervision:**
The central bank supervises and regulates commercial banks and other financial institutions to ensure they operate within legal and regulatory frameworks.

8. Distinguish between 'Macroprudential' and 'Microprudential' regulations. 97(1.c)

Distinction between Macroprudential and Microprudential Regulations:

Aspect	Macroprudential Regulations	Microprudential Regulations
Focus	Focuses on the stability of the entire financial system .	Focuses on the stability of individual financial institutions .
Objective	Aims to prevent systemic risk and protect the overall economy from shocks.	Aims to ensure the soundness and stability of individual institutions.
Scope	Deals with issues that could affect the broader economy (e.g., asset bubbles, interconnections).	Deals with issues related to individual institutions (e.g., capital adequacy, liquidity).
Risk Management	Focuses on mitigating risks that could lead to financial crises.	Focuses on managing risks within a specific institution.
Examples of Tools	Capital surcharges, liquidity requirements, stress testing of the financial system.	Capital requirements, prudential ratios, supervisory interventions.

Short notes:

Green Banking

Green Banking refers to environmentally responsible banking practices that promote sustainable development. In Bangladesh, the Bangladesh Bank has issued guidelines to encourage banks to adopt eco-friendly policies, such as financing renewable energy projects, paperless banking, and green loans. Banks like DBBL and BRAC Bank have introduced green products to reduce environmental impact. Green Banking helps reduce carbon footprints, supports eco-friendly industries, and ensures financial stability while promoting sustainability in Bangladesh.

Inclusive Finance

Inclusive Finance refers to financial services that ensure access to banking and financial products for all, especially underserved and low-income groups. In Bangladesh, initiatives like mobile banking (bKash, Nagad), agent banking, and microfinance programs (Grameen Bank, BRAC) have improved financial inclusion. The Bangladesh Bank has taken steps to promote SME financing, rural banking, and women entrepreneurship. Inclusive Finance helps reduce poverty, promotes economic growth, and ensures financial stability by integrating all segments of society into the financial system.

Sustainable Finance

Sustainable Finance refers to financial activities that support economic growth while ensuring environmental and social sustainability. In Bangladesh, banks and financial institutions are encouraged by Bangladesh Bank to invest in green projects, renewable energy, and social enterprises. Initiatives like Green Bonds and Sustainability-Linked Loans promote long-term sustainability. Sustainable Finance helps reduce environmental risks, supports climate-friendly investments, and ensures financial stability while fostering sustainable economic development.

Financial Instruments

Financial instruments are assets that can be traded in financial markets, such as stocks, bonds, and derivatives. In Bangladesh, common financial instruments include Treasury bills, commercial papers, and corporate bonds. These instruments help in investment, risk management, and liquidity management, playing a crucial role in the country's financial system.

Financial Intermediaries

Financial intermediaries are institutions that connect savers and borrowers, facilitating the flow of funds. In Bangladesh, banks, non-bank financial institutions (NBFIs), insurance companies,

and microfinance institutions act as intermediaries. They help in economic development by providing loans, investment options, and financial services.

Financial Market

The financial market is a platform where financial instruments are traded. In Bangladesh, the financial market consists of the money market (short-term funds) and the capital market (long-term investments). The Dhaka Stock Exchange (DSE) and Chittagong Stock Exchange (CSE) are key components of the capital market, while the Bangladesh Bank regulates the money market.

Repurchase Agreements

Repurchase agreements (Repo) are short-term borrowing agreements where securities are sold with a commitment to repurchase them later. In Bangladesh, banks and financial institutions use repos for short-term liquidity management under Bangladesh Bank regulations. It helps stabilize the money market by ensuring adequate liquidity.

Corporate Bond

A corporate bond is a debt instrument issued by companies to raise capital. In Bangladesh, corporate bonds are gaining popularity as an alternative financing option. These bonds help businesses expand operations while providing investors with fixed-income returns. Bangladesh Securities and Exchange Commission (BSEC) regulates corporate bonds in the country.

Stock Market

The stock market is a marketplace where shares of publicly listed companies are traded. In Bangladesh, the Dhaka Stock Exchange (DSE) and Chittagong Stock Exchange (CSE) facilitate stock trading. It provides companies with capital and investors with opportunities to earn returns, playing a vital role in economic growth.

Fixed Exchange Rates

A fixed exchange rate system is where a country's currency value is tied to another currency, such as the US dollar. Bangladesh follows a managed exchange rate system, where Bangladesh Bank intervenes to maintain currency stability. Fixed exchange rates help in trade stability but may require high foreign reserves to maintain.

Freely Floating Exchange Rate System

A freely floating exchange rate system allows currency value to be determined by market forces of demand and supply. Bangladesh does not follow a fully floating system but rather a managed floating system. A floating rate system provides flexibility but can lead to currency fluctuations.

Pegged Exchange Rate System

A pegged exchange rate system links a country's currency to another currency or a basket of currencies. Bangladesh maintains a managed exchange rate where the Taka is influenced by the US dollar. Pegged systems provide stability in trade but require central bank intervention to maintain the rate.

Lender of the Last Resort

A lender of last resort is an institution, usually a central bank, that provides emergency funds to financial institutions facing liquidity crises. In Bangladesh, Bangladesh Bank acts as the lender of last resort, ensuring financial system stability by supporting banks during financial distress.

Clearing House Operations

Clearing house operations involve processing and settling financial transactions between banks. In Bangladesh, Bangladesh Automated Clearing House (BACH) manages electronic cheque clearing and fund transfers. It ensures smooth financial transactions and reduces settlement risks.

Bai Muazzal

Bai Muazzal is an Islamic banking mode where banks provide goods or assets on deferred payment. In Bangladesh, Islamic banks use Bai Muazzal for Shariah-compliant financing, allowing customers to purchase goods with payment in installments. It is widely used for trade and business financing.

Banker's Bank

A banker's bank refers to a central bank that regulates and supervises commercial banks. In Bangladesh, Bangladesh Bank is the banker's bank, providing liquidity support, monetary policy regulation, and financial oversight. It ensures the stability and efficiency of the banking system.

Mathematical Problem

Problem-01:

Imagine that an economy has the following monetary aggregates at the end of 31-12-2023:

Currency Outside Bank (COB)	M1	M2
Tk. 85,000	Tk. 2,35,000	Tk. 7,45,000

On 01-01-2024, the following transactions were executed:

- Encashment of an FDR worth Tk. 1,00,000 of that Sanchayapatra worth Tk. 50,000 was purchased from BB and rest of the Tk. 50,000 was deposited in a current account.
- Withdrawal of Tk. 15,000 from a savings account and kept that in cash.
- Cash out transaction of Tk. 20,000 with an MFS agent. [MFS maintains a trust and settlement account with a scheduled bank.]

Calculate COB, M1, and M2 monetary aggregates at the end of 01-01-2024.

COB, M1 and M2 Calculation:

Transactions on 01-01-2024:

(i)

Currency Outside Bank (COB)	Tk. 85,000
M1	Tk. 2,35,000
M2	Tk. 7,45,000

Encashment of an FDR worth Tk. 1,00,000 of that Sanchayapatra worth Tk. 50,000 was purchased from BB and rest of the Tk. 50,000 was deposited in a current account.

- Encashment of an FDR worth Tk. 1,00,000 (Decreasing M2 by Tk. 1,00,000)
- Tk. 50,000 used to purchase Sanchayapatra (Increasing M2 by Tk. 50,000)
- Tk. 50,000 deposited into a current account (Increasing M1 and M2 as well by Tk. 50,000)

Effects on Monetary Aggregates:

$$\text{COB} = \text{Tk. 85,000}$$

$$\text{M1} = \text{Tk. 2,35,000} + \text{Tk. 50,000} = \text{Tk. 2,85,000}$$

$$\text{M2} = \text{Tk. 7,45,000} - \text{Tk. 1,00,000} + \text{Tk. 50,000} + \text{Tk. 50,000} = \text{Tk. 7,45,000}$$

(ii)

Currency Outside Bank (COB)	Tk. 85,000
M1	Tk. 2,85,000
M2	Tk. 7,45,000

Withdrawal of Tk. 15,000 from a savings account and kept that in cash.

- Withdrawal of Tk. 15,000 from a savings account (Decreasing M2 by Tk. 15,000)
- kept that in cash (Increasing COB, M1 and M2 by Tk. 15,000)

Effects on Monetary Aggregates:

$$\begin{array}{lll} \text{COB} & = \text{Tk. 85,000} + \text{Tk. 15,000} & = \text{Tk. 1,00,000} \\ \text{M1} & = \text{Tk. 2,85,000} + \text{Tk. 15,000} & = \text{Tk. 3,00,000} \\ \text{M2} & = \text{Tk. 7,45,000} - \text{Tk. 15,000} + \text{Tk. 15,000} & = \text{Tk. 7,45,000} \end{array}$$

(iii)

Cash out transaction of Tk. 20,000 with an MFS agent. [MFS maintains a trust and settlement account with a scheduled bank.]

- Cash-out transaction of Tk. 20,000 with an MFS agent (Decreasing M2 by Tk. 20,000)
- Cash (Increasing COB, M1 and M2 by Tk. 20,000)

Currency Outside Bank (COB)	Tk. 1,00,000
M1	Tk. 3,00,000
M2	Tk. 7,45,000

Effects on Monetary Aggregates:

$$\begin{array}{lll} \text{COB} & = \text{Tk. 1,00,000} + \text{Tk. 20,000} & = \text{Tk. 1,20,000} \\ \text{M1} & = \text{Tk. 3,00,000} + \text{Tk. 20,000} & = \text{Tk. 3,20,000} \\ \text{M2} & = \text{Tk. 7,45,000} - \text{Tk. 20,000} + \text{Tk. 20,000} & = \text{Tk. 7,45,000} \end{array}$$

Problem-02:

From the following data of Bangladesh calculate “Monetization Ratio” and “New Issue Ratio” for FY 2023 and interpret them :

Year	GNP at current market prices	Money supply (M ₂)	Assets of all FIs
FY 2022	41,291	17,081	21,349
FY 2023	46,185	18,872	24,140

Solution:

Monetization Ratio and New Issue Ratio Calculation for Bangladesh (FY 2023)

Formulas:

1. Monetization Ratio (MR) = $\frac{M_2}{GNP} \times 100$
2. New Issue Ratio (NIR) = $\frac{\Delta M_2}{\text{Total Financial Assets (Current Year)}} \times 100$

Calculations for FY 2023:

1. Monetization Ratio (MR)

- Given:
 - M₂ = 18,872 (in billion BDT)
 - GNP = 46,185 (in billion BDT)

$$MR = \left(\frac{18,872}{46,185} \right) \times 100 \approx 40.86\%$$

2. New Issue Ratio (NIR)

- Given:
 - M₂ in FY 2023 = 18,872
 - M₂ in FY 2022 = 17,081
 - Total Financial Assets (Current Year) = 24,140 (in billion BDT)

- Calculate $\Delta M2$:

$$\Delta M2 = M2_{FY2023} - M2_{FY2022} = 18,872 - 17,081 = 1,791$$

- Calculate NIR:

$$NIR = \left(\frac{1,791}{24,140} \right) \times 100 \approx 7.42\%$$

Interpretation:

- **Monetization Ratio (40.86%):** This indicates that 40.86% of Bangladesh's GNP is represented by the money supply (M2), reflecting the level of liquidity in the economy. A higher ratio suggests greater monetization, implying improved financial intermediation.
- **New Issue Ratio (7.42%):** This shows the proportion of newly created money supply relative to total financial assets. A higher NIR indicates a more dynamic financial system with increased money issuance.

These indicators help assess financial deepening and liquidity trends in Bangladesh's economy for FY 2023.

Problem-03:

Find out the price of a two-year, 10% coupon bond (semi-annual coupon payments) with a face value of Tk. 1,000 and a yield to maturity 12%.

Solution:

Bond Price Calculation (PV):

Formula:

$$\text{Yield to Maturity (YTM)} = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

Face Value , FV = Tk. 1000

Coupon Payment = 10%

Semi-annual Coupon Payment, C = 5% of 1000 = Tk. 50

Yield to Maturity, YTM = 12% (annually)

YTM = 6% (Semi-annual)

Number of Period, n = 4 (1 Year = 2 periods)

Bond Price Calculation (PV):

$$\text{YTM} = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

$$\Rightarrow 0.06 = \frac{50 + \frac{1000 - PV}{4}}{\frac{1000 + PV}{2}}$$

$$\Rightarrow \frac{6}{100} * \frac{1000 + PV}{2} = \frac{200 + 1000 - PV}{4}$$

$$\Rightarrow \frac{6}{100} * \frac{1,000 + PV}{2} = \frac{200 + 1,000 - PV}{4}$$

$$\Rightarrow \frac{3,000 + 3PV}{100} = \frac{1,200 - PV}{4}$$

$$\Rightarrow 12,000 + 12 PV = 1,20,000 - 100 PV$$

$$\Rightarrow 112 PV = 1,08,000$$

$$\Rightarrow PV = \frac{1,08,000}{112}$$

$$\therefore PV = 964.29$$

Problem-04:

You decided to buy a stock that is currently selling for Tk. 55 per share and pays dividend Tk. 4 per year. The market analyst predicts that the stock will be sold at Tk. 60 in one year. Your expected rate of return is 12% :

- (i) Calculate the current price of the stock.
- (ii) Should you buy this stock? Why the stock is selling for less than market price?

Solution:

Bond Price Calculation (P_0):

Formula:

$$\text{Stock Price, } P_0 = \frac{D_1 + P_1}{1 + r}$$

Given,

Dividend per year, $D_1 = \text{Tk. } 4$

Expected Rate of Return, $r = 12\% = 0.12$

Expected stock price, $P_1 = \text{Tk. } 60$

$$\begin{aligned} \text{Stock Price, } P_0 &= (4 + 60) / (1 + 0.12) \\ &= 64 / 1.12 \\ &= \text{Tk. } 57.14 \end{aligned}$$

Problem-05:

Suppose the following table shows different forms of money of Bangladesh Economy:

(Billion of Tk.)

	2020	2021	2022
Currency Outside Bank	1840	1850	1865
Demand deposit	1944	1960	1986
Current deposit	1362	1358	1376
Savings deposit	5780	5968	6105
Time deposit	4212	4794	5790

(i) Calculate M_1 and M_2 money supply for each year and growth rate of M_1 and M_2 from the previous year.

Solution:

Step 1: M_1 and M_2 Calculation Formulas

- M_1 (Narrow Money) = Currency Outside Bank + Demand Deposits + Other Checkable Deposits (Current Deposits)

$$M_1 = \text{Currency Outside Bank} + \text{Demand Deposit} + \text{Current Deposit}$$

- M_2 (Broad Money) = M_1 + Savings Deposits + Time Deposits

$$M_2 = M_1 + \text{Savings Deposit} + \text{Time Deposit}$$

Step 2: Calculate M_1 and M_2 for Each Year

For 2020:

$$M_1 = 1840 + 1944 + 1362 = 5146$$

$$M_2 = 5146 + 5780 + 4212 = 15138$$

For 2021:

$$M_1 = 1850 + 1960 + 1358 = 5168$$

$$M_2 = 5168 + 5968 + 4794 = 15930$$

For 2022:

$$M_1 = 1865 + 1986 + 1376 = 5227$$

$$M_2 = 5227 + 6105 + 5790 = 17122$$

Step 3: Calculate Growth Rates

Growth Rate Formula:

$$\text{Growth Rate} = \left(\frac{\text{Current Year Value} - \text{Previous Year Value}}{\text{Previous Year Value}} \right) \times 100$$

M₁ Growth Rate

$$\text{Growth Rate of } M_1(2021) = \left(\frac{5168 - 5146}{5146} \right) \times 100 = 0.43\%$$

$$\text{Growth Rate of } M_1(2022) = \left(\frac{5227 - 5168}{5168} \right) \times 100 = 1.14\%$$

M₂ Growth Rate

$$\text{Growth Rate of } M_2(2021) = \left(\frac{15930 - 15138}{15138} \right) \times 100 = 5.24\%$$

$$\text{Growth Rate of } M_2(2022) = \left(\frac{17122 - 15930}{15930} \right) \times 100 = 7.48\%$$

Final Answer

Year	M ₁ (Billion Tk.)	M ₂ (Billion Tk.)	M ₁ Growth Rate (%)	M ₂ Growth Rate (%)
2020	5146	15138	-	-
2021	5168	15930	0.43%	5.24%
2022	5227	17122	1.14%	7.48%

Problem-06:

Calculate the present value of a taka 1000 zero coupon bond with 7 years to maturity if the yield to maturity is 8%.

Solution:

$$PV = \frac{FV}{(1 + r)^N}$$

Where:

- FV = Face Value = BDT 1,000
 - r = Yield to Maturity (YTM) = 8% or 0.08
 - N = Number of years to maturity = 7
-

1. Using the Standard Formula:

$$\begin{aligned} PV &= \frac{1000}{(1 + 0.08)^7} \\ &= \frac{1000}{1.08^7} \\ &= \frac{1000}{1.713} \\ &= 583.49 \end{aligned}$$

Problem-07:

Calculate the yield when a bank purchase a T-bill with six month maturity and Tk. 1,000 par value for 960.

Solution:

$$\text{Discount Yield (DY)} = \left(\frac{\text{Face Value} - \text{Purchase Price}}{\text{Purchase Price}} \right) \times \frac{365}{\text{Days to Maturity}}$$

Given Data:

- Face Value (FV) = BDT 1,000
 - Purchase Price (PP) = BDT 960
 - Maturity = 6 months = 182 days
-

Calculation:

$$\begin{aligned}
 DY &= \left(\frac{1000 - 960}{960} \right) \times \frac{365}{182} \\
 &= \left(\frac{40}{960} \right) \times 2 \\
 &= 0.04167 \times 2 \\
 &= 0.0833 \text{ or } 8.33\%
 \end{aligned}$$

Problem-08:

From the following list of items, indicate if they are in M1, M2 or neither:

SL	Items	Amount in BDT
1	Line of credit with your bank	50,000
2	Money in your pocket	10,000
3	Money in your checking account	20,000
4	An FDR Issue from your bank	50,000
5	Value of Travelers check in you have	1,500
6	A Sanchay Patra is purchased from Bangladesh Bank	1,00,000

Solution:

Here's the classification of each item as M1, M2, or neither:

Serial No.	Items	Amount (BDT)	M1 / M2 / Neither
1	Line of credit with your bank	50,000	Neither
2	Money in your pocket	10,000	M1
3	Money in your checking account	20,000	M1
4	An FDR issued from your bank	50,000	M2
5	Value of traveller's check you have	1,500	M1
6	A sanchayapatra purchased from Bangladesh Bank	100,000	M2

Explanation:

- **M1:** Includes very liquid forms of money, such as cash, checking accounts, and traveller's checks.
- **M2:** Includes everything in M1 plus less liquid assets, such as fixed deposits (FDR) and sanchayapatras (savings certificates).
- **Line of credit:** This is a form of loan and not considered liquid cash, so it is classified as "neither."

Problem-09:

Calculate the amount of the time deposits when narrow money of around BDT 450 crore is 30% of broad money as on 30 June 2023. Calculate the amount of currency outside banks if demand deposit is almost 60% of M1 on the same date.

Solution:

Part 1: Calculate the Amount of Time Deposits (M2)

1. Given:

- Narrow money (M1) = BDT 450 crore
- M1 is 30% of broad money (M2)

2. Find M2:

$$M1 = 0.30 \times M2 \implies M2 = \frac{M1}{0.30}$$

$$M2 = \frac{450 \text{ crore}}{0.30} = 1500 \text{ crore}$$

3. Calculate Time Deposits (M2 = M1 + Time Deposits):

$$\text{Time Deposits} = M2 - M1$$

$$\text{Time Deposits} = 1500 \text{ crore} - 450 \text{ crore} = 1050 \text{ crore}$$

Part 2: Calculate the Amount of Currency Outside Banks

1. Given:

- Demand deposits are 60% of M1.
- M1 = BDT 450 crore.

2. Calculate Demand Deposits:

$$\text{Demand Deposits} = 0.60 \times M1 = 0.60 \times 450 \text{ crore} = 270 \text{ crore}$$

3. Calculate Currency Outside Banks:

$$\text{Currency Outside Banks} = M1 - \text{Demand Deposits}$$

$$\text{Currency Outside Banks} = 450 \text{ crore} - 270 \text{ crore} = 180 \text{ crore}$$

Problem-10:

An investors purchased a 6 months (182 days) T-bill at a price of BDT 9500 which face value is BDT 10,000, If the T-bill is HTM, calculate its yield. 6 (b) What is amortized yield of the bill in part (a) if it is sold after holding for 100 days at a price of BDT 9,800?

Solution:

$$\text{Discount Yield (DY)} = \left(\frac{\text{Face Value} - \text{Purchase Price}}{\text{Purchase Price}} \right) \times \frac{365}{\text{Days to Maturity}}$$

Given Data:

- Face Value (FV) = BDT 10,000
- Purchase Price (PP) = BDT 9,500
- Days to Maturity (D) = 182 days

Calculation:

$$\begin{aligned} \text{DY} &= \left(\frac{10,000 - 9,500}{9,500} \right) \times \frac{365}{182} \\ &= \left(\frac{500}{9,500} \right) \times 2 \\ &= 0.0526 \times 2 \\ &= 0.1052 \text{ or } 10.52\% \end{aligned}$$

Problem-11:

What is amortized yield of the bill in part (a) if it is sold after holding for 100 days at a price of BDT 9,800?

Solution:**Holding Yield Formula:**

$$\text{HPY} = \left(\frac{\text{Selling Price} - \text{Purchase Price}}{\text{Purchase Price}} \right) \times \frac{365}{\text{Holding Period}}$$

Given Data:

- Purchase Price (PP) = BDT 9,500
- Selling Price (SP) = BDT 9,800
- Holding Period (HP) = 100 days

Calculation:

$$\begin{aligned}
 \text{HPY} &= \left(\frac{9,800 - 9,500}{9,500} \right) \times \frac{365}{100} \\
 &= \left(\frac{300}{9,500} \right) \times 3.65 \\
 &= 0.0316 \times 3.65 \\
 &= 0.1153 \text{ or } 11.53\%
 \end{aligned}$$

Problem-12:

If a bank purchase 100 units of 90 days commercial papers with a par value of BDT 10,000 at a price of BDT 9,85

Solution:

Discount Yield,

$$\text{DY} = \left(\frac{\text{Face Value} - \text{Purchase Price}}{\text{Purchase Price}} \right) \times \frac{365}{\text{Days to Maturity}}$$

Given Data:

- Face Value (FV) = BDT 10,000
- Purchase Price (PP) = BDT 9,850
- Days to Maturity (D) = 90 days

Calculation:

$$\begin{aligned}
 \text{DY} &= \left(\frac{10,000 - 9,850}{9,850} \right) \times \frac{365}{90} \\
 &= \left(\frac{150}{9,850} \right) \times 4.056 \\
 &= 0.0152 \times 4.056 \\
 &= 0.0616 \text{ or } 6.16\%
 \end{aligned}$$

2. Calculating the Total Profit:

Since the bank purchased 100 units, the profit per unit and the total profit are:

$$\text{Profit per Unit} = \text{Face Value} - \text{Purchase Price} = 10,000 - 9,850 = \text{BDT}150$$

$$\text{Total Profit} = 100 \times 150 = \text{BDT}15,000$$

Problem-13:

From the following information compute the financial account balance:

Items	Amount in BDT
Net Foreign Direct Investment	66,000
Net Portfolio Investment	(40,000)
Net Wage earners remittance	50,000
Asset Funding	30,000
Errors and Omissions	10,000

Solution:

To calculate the Financial Account Balance, we consider only the components that are part of the financial account, excluding current account items like remittances and errors/omissions.

Financial Account Components:

1. **Net Foreign Direct Investment (FDI)** = BDT 66,000 bn
2. **Net Portfolio Investment** = BDT (40,000 bn) *(negative indicates an outflow)*
3. **Asset Funding** = BDT 30,000

Calculation of Financial Account Balance:

$$\begin{aligned}\text{Financial Account Balance} &= \text{Net FDI} + \text{Net Portfolio Investment} + \text{Asset Funding} \\ &= 66,000 + (-40,000) + 30,000 \\ &= 66,000 - 40,000 + 30,000 \\ &= 56,000 \text{ bn BDT}\end{aligned}$$

Problem-14:

On 01 January 2023, company A issues Tk. 1000 par value with annual coupon payment of Tk. 75 and bond will be matured on 31 December 2025. If after the issuance of the bond the yield to maturity (YTM) of bond stands 9%, calculate the current market price of the bond.

Solution:

Bond Price Calculation (PV):

$$\text{Yield to Maturity (YTM)} = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

Given,

Coupon Payment, $C = \text{Tk. } 75$

Face Value, $FV = \text{Tk. } 1000$

Yield to Maturity, $\text{YTM} = 9\%$

Number of Period, $n = 3$

Current Market price of the bond, $PV = ?$

$$\text{YTM} = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

$$\Rightarrow 0.09 = \frac{75 + \frac{1000 - PV}{3}}{\frac{1000 + PV}{2}}$$

$$\Rightarrow \frac{9}{100} * \frac{1000 + PV}{2} = \frac{225 + 1000 - PV}{3}$$

$$\Rightarrow \frac{9}{100} * \frac{1,000 + PV}{2} = \frac{225 + 1,000 - PV}{3}$$

$$\Rightarrow \frac{9,000 + 9PV}{200} = \frac{1,225 - PV}{3}$$

$$\Rightarrow 27,000 + 27 PV = 2,45,000 - 200 PV$$

$$\Rightarrow 227 PV = 2,18,000$$

$$\Rightarrow PV = \frac{2,18,000}{227}$$

$$\therefore PV = 960.35$$

Problem-15:

What is the selling price of a stock if its next year dividends are expected to be Tk. 3.5 while the discount rate is estimated to be 9% and the dividends are projected to increase at 5% per year indefinitely (Please use the Gordon growth model)?

Solution:

$$P_0 = \frac{D_1}{r - g}$$

Where:

- P_0 = Current stock price (selling price)
 - D_1 = Expected dividend next year = BDT 3.5
 - r = Discount rate (required rate of return) = 9% or 0.09
 - g = Growth rate of dividends = 5% or 0.05
-

Calculation:

$$\begin{aligned} P_0 &= \frac{3.5}{0.09 - 0.05} \\ &= \frac{3.5}{0.04} \\ &= 87.50 \end{aligned}$$

Problem-16:

From the following information measures the monetary aggregates of M1 and M2:

Items	Value as of December, 2022 (billion TK)
Currency in circulation	2923.60
Currency in tills of DMBs	241.80
Demand Deposits	1837.40
Other checkable Deposits	6.20
Time Deposits	13054.30
Money Market Mutual Fund	10.50

Solution:

Calculation of Monetary Aggregates (M1 and M2)

Formula for M1,

$$M1 = \text{Currency in Circulation} + \text{Demand Deposits} + \text{Other Checkable Deposits}$$

Calculation of Currency in Circulation:

$$\text{Currency in Circulation} = \text{Total Currency Issued} - \text{Currency in Tills of DMBs}$$

$$= 2923.60 - 241.80$$

$$= 2681.80 \text{ billion taka}$$

$$M1 = 2681.80 + 1837.40 + 6.20$$

$$= 4525.40 \text{ billion taka}$$

Formula for M2:

$$M2 = M1 + \text{Time Deposits} + \text{Money Market Mutual Funds}$$

$$M2 = 4525.40 + 13054.30 + 10.50$$

$$= 17590.20 \text{ billion taka}$$

Problem-17:

What is the discount yield for a five year bond that was purchased for Tk. 60 and has a face value of Tk. 100.

Solution:

Yield to Maturity (YTM) Calculation Using the Given Formula

Formula:

$$YTM = \left(\frac{\text{Discount}}{\text{Face Value}} \right) * \left(\frac{12}{\text{Number of Months to Maturity}} \right) * 100$$

Given Data:

- Face Value (FV) = Tk. 100
- Purchase Price (P) = Tk. 60
- Discount = Face Value - Purchase Price = 100 - 60 = Tk. 40
- Number of Months to Maturity = 5 years = 5 * 12 = 60 months

Calculation:

$$\begin{aligned} YTM &= \left(\frac{40}{100} \right) * \left(\frac{12}{60} \right) * 100 \\ &= 0.4 * 0.2 * 100 \\ &= 0.08 * 100 = 8\% \end{aligned}$$

Problem-18:

The average industry PE ratio for hotels similar to Shangri-La hotel Chain is 37. What is the current price of Shangri-La's earnings per share (EPS) are projected to be Tk. 1.16?

Solution:

Calculation of Current Price Using P/E Ratio

Formula:

$$\text{Current Price} = \text{EPS} \times \text{P/E Ratio}$$

Given Data:

- Average Industry P/E Ratio = 37
- Projected EPS = Tk. 1.16

Calculation:

$$\begin{aligned} \text{Current Price} &= 1.16 \times 37 \\ &= 42.92 \end{aligned}$$